

Western University
OF HEALTH SCIENCES

College of
Dental Medicine

Materials

A PRECLINICAL AND CLINICAL MATERIAL REFERENCE GUIDE

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This document is meant to be used as a reference guide for the various materials throughout Western University of Health Sciences, Center for Dental Medicine. It is also meant to keep the materials throughout the clinic consistent and will be regularly updated with any new material changes.

As a reference guide, it should not and does not replace using the manufacturers' inserts for each material when questions, confusion, or discrepancies arise. If something is found to be unclear or incorrect, please bring it to the attention of Dr. Flask for editing and revision.

Pictures found throughout the document largely came from manufacturers' websites or brochures. They are not meant for any advertising purposes, but only to help students know the material they are searching for. If there are any infringements, it was not done on purpose and will be rectified to all parties' satisfaction once attention is brought to the instigating content.

No financial considerations were taken into account in the fabrication of this manual.

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Anesthetics

Anesthetics are a vital part of the dental armamentarium. From Nitrous Oxide being used in 1779 to Ethyl chloride for dental patients in 1894, anesthetics have been a part of dental procedures for hundreds of years. Later, the notorious Procaine (Novocaine) was discovered in 1904, and later still, Lidocaine in 1943 in minor dental surgeries which is still commonly used today.

The key to proper administration of local anesthetic is through administering a therapeutic dose, or smallest effective dose, in order to avoid adverse reactions such as local toxicity, allergies, cardiovascular influences, and drug interactions.

True allergies to anesthetics are less than 1% of all adverse reactions to anesthetics; however, they are a common concern for patients. This is likely partially derived from a historical hypersensitivity to ester-linked anesthetics such as procaine which contain PABA. While topical anesthetics are still commonly esters, the local injectable anesthetics are more commonly of the amide variety, believed to have a significantly lower risk of true allergic reactions.

1. Topical Anesthetics

As discussed, some topical anesthetics still commonly utilize ester-linked anesthetics. This should be taken into consideration before applying a topical anesthetic to a patient with sensitivity to PABA.

a. ***LolliCaine*** *(Centrix Dental)*

- This product is a fast-acting 20% benzocaine gel in a convenient single use package.



Instructions for Use

1. Remove one LolliCaine unit
2. Slowly peel back the lid, exposing the cotton swab applicator and unit dose well
3. Dib the swab applicator and apply gel to mucosa or gingival soft tissue
4. Wait 1 minute or until patient feels numb before starting local injection

---

b. ***Oraqix*** *(Dentsply)*

- Lidocaine (2.5%) and Prilocaine (2.5%) gel (both amide anesthetics) is primarily used on the periodontium for quick pain relief during SRP type procedures.
- This product is **NOT** for injection.
- Duration: **20 minutes**
- One cartridge = 1.7g, and will usually cover one quadrant



Instructions for Use

1. Apply around the desired gingival margin using the corresponding blunt-tipped applicator
2. Wait 30 seconds
3. Fill the periodontal pockets with Oraqix, again using the blunt-tipped applicator until the gel becomes visible at the gingival margin
4. Wait 30 seconds
5. Begin treatment

---

c. ***Cetacaine Liquid Topical Anesthetic (14%)*** *(Cetylite)*

- Cetacaine's active ingredients include: Benzocaine (14%), Butamben (2.0%), and Tetracaine Hydrochloride (2.0%)



Instructions for Use

1. Dispense a drop of product in a dappen dish
2. Apply 200mg (0.2mL) directly to the tissue using a microbrush
   - Excess of 400mg (0.4mL) is contraindicated
3. Continue with care

---

d. ***HurriCaine Spray (20%)*** *(Beutlich)*

- HurriCaine spray is a fast acting 20% Benzocaine Oral Anesthetic
- Anesthesia is accomplished in about 20sec



Instructions for Use

1. Insert extension tube and position 1-2inches from affected area
2. Spray ½sec (press and immediately release actuator)
3. Repeat if necessary

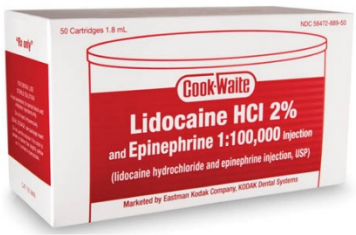
2. Injectable Local Anesthetics

- Calculations for dosing anesthetics are done based on a patient's body weight. Milligrams of medication per kilogram of the patient.
- There has historically been a changing environment in the maximum recommended dosing (MRD) for local injectable anesthetics. Though the FDA and manufacturer recommendations may allow for higher levels than listed below, students should never need to exceed the listed maximums. Often, the listed maximums are for a healthy adult patient, while the below values come from pediatric calculations.
- If a patient is not sufficiently anesthetized after **2 carpules**, faculty should be made aware and potentially intervene.

---

*a. **Lidocaine HCl 2% and Epinephrine 1:100,000** (Henry Schein)*

- 20mg Lidocaine to 0.01mg Epinephrine to every 1.0mL
- In a 1.8mL carpule: 36mg Lidocaine to 0.018mg Epinephrine
- Maximum recommended dose is 4.4mg/kg or 2.0mg/lb not to exceed 300mg in pediatric cases.
- Lidocaine is a Pregnancy **Category B** drug, and can therefore be used during pregnancy



Patient Weight	Maximum Carpules	Lidocaine:Epinephrine
23kg or 50lbs	2.8 Carpules	101mg:0.05mg
45kg or 100lbs	5.5 Carpules	198mg:0.099mg
68kg or 150lbs	8.3 Carpules	300mg:0.15mg
90kgs or 200lbs	8.3 Carpules	300mg:0.15mg

Time of Onset	Duration at Pulp	Duration at Tissue
5 to 10min	60 to 85min	170 to 190min

---

*b. **Lidocaine HCl 2% and Epinephrine 1:50,000** (Henry Schein)*

- 20mg Lidocaine to 0.02mg Epinephrine to every 1.0mL
- In a 1.8mL carpule: 36mg Lidocaine to 0.036mg Epinephrine
- Maximum recommended dose is 4.4mg/kg or 2.0mg/lb not to exceed 300mg in pediatric cases.
- Lidocaine is a Pregnancy **Category B** drug, and can therefore be used during pregnancy



Patient Weight	Maximum Carpules	Lidocaine:Epinephrine
23kg or 50lbs	2.8 Carpules	101mg:0.056mg
45kg or 100lbs	5.5 Carpules	198mg:0.110mg
68kg or 150lbs	8.3 Carpules	300mg:0.166mg
90kgs or 200lbs	8.3 Carpules	300mg:0.166mg

Time of Onset	Duration at Pulp	Duration at Tissue
5 to 10min	60 to 85min	170 to 190min

c. ***Marcaine 0.5% and Epinephrine 1:200,000*** *(Cook-Waite)*

- 5mg Marcaine to 0.005mg Epinephrine to every 1.0mL
- In a 1.8mL carpule: 9mg Marcaine to 0.009mg Epinephrine
- Maximum recommended dose is 1.3mg/kg or 0.6mg/lb not to exceed 90mg in pediatric cases
- Marcaine is also not recommended for patients younger than 12 years old
- Marcaine is a Pregnancy **Category C** drug, and therefore should NOT be used during pregnancy



Patient Weight	Maximum Carpules	Marcaine:Epinephrine
23kg or 50lbs	3.3 Carpules	29.9mg:0.030mg
45kg or 100lbs	6.5 Carpules	58.5mg:0.059mg
68kg or 150lbs	9.8 Carpules	88.4mg:0.088mg
90kgs or 200lbs	10 Carpules	90mg:0.09mg

Time of Onset	Duration at Pulp	Duration at Tissue
2 to 10min	40 to 240min	340 to 440min

---

d. ***Septocaine (Articaine Hydrochloride 4% with Epinephrine 1:100,000)*** *(Septodent)*

- 40mg Articaine to 0.01mg Epinephrine to every 1.0mL
- In a 1.8mL carpule: 72mg Articaine to 0.018mg Epinephrine
- Maximum recommended dose is 7.0mg/kg or 3.2mg/lb not to exceed 500mg in pediatric cases
- Articaine is a Pregnancy **Category C** drug, and therefore should NOT be used during pregnancy
- Articaine has had a history of research indicating increased risk of nerve damage when administered to the IAN. This evidence is fairly anecdotal and weak, with the drug being cleared by the FDA for nerve block anesthesia. Despite this, the use of Articaine must be approved by the attending faculty and blocking the IAN should be avoided.



Patient Weight	Maximum Carpules	Mepivacaine
23kg or 50lbs	2.2 Carpules	161mg:0.04mg
45kg or 100lbs	4.4 Carpules	315mg:0.079mg
68kg or 150lbs	6.6 Carpules	476mg:0.119mg
90kgs or 200lbs	6.9 Carpules	500mg:0.125mg

Time of Onset	Duration at Pulp	Duration at Tissue
8 to 10min	60 to 90min	190 to 230min

---

e. ***Mepivacaine HCl 3% without vasoconstrictor*** *(Henry Schein)*

- 30mg Mepivacaine to every 1.0mL
- In a 1.8mL carpule: 54mg Mepivacaine
- Maximum recommended dose is 4.4mg/kg or 2.0mg/lb not to exceed 300mg in pediatric cases
- Mepivacaine is a Pregnancy **Category C** drug, and therefore should NOT be used during pregnancy



Patient Weight	Maximum Carpules	Mepivacaine
23kg or 50lbs	1.9 Carpules	101mg
45kg or 100lbs	3.7 Carpules	198mg
68kg or 150lbs	5.5 Carpules	300mg
90kgs or 200lbs	5.5 Carpules	300mg

Time of Onset	Duration at Pulp	Duration at Tissue
3 to 5min	25 to 40min	90 to 165min

Table 1: Condensed Chart of Anesthetics

	Patient Weight	Maximum Carpules	Anesthetic to Epinephrine Ratio		Time of Onset	Duration at Pulpal	Duration at Tissue
Lidocaine 2% with 1:100,000 Epinephrine	23kg or 50lbs	2.8 Carpules	101mg:0.05mg		5 to 10min	60 to 85min	170 to 190min
	45kg or 100lbs	5.5 Carpules	198mg:0.099mg				
	68kg or 150lbs	8.3 Carpules	300mg:0.15mg				
	90kgs or 200lbs	8.3 Carpules	300mg:0.15mg				
Lidocaine 2% with 1:50,000 Epinephrine	23kg or 50lbs	2.8 Carpules	101mg:0.056mg		5 to 10min	60 to 85min	170 to 190min
	45kg or 100lbs	5.5 Carpules	198mg:0.110mg				
	68kg or 150lbs	8.3 Carpules	300mg:0.166mg				
	90kgs or 200lbs	8.3 Carpules	300mg:0.166mg				
Mepivicaine 3%	23kg or 50lbs	1.9 Carpules	101mg		3 to 5min	25 to 40min	90 to 165min
	45kg or 100lbs	3.7 Carpules	198mg				
	68kg or 150lbs	5.5 Carpules	300mg				
	90kgs or 200lbs	5.5 Carpules	300mg				
Marcaine 0.5% with 1:200,000 Epinephrine	23kg or 50lbs	3.3 Carpules	29.9mg:0.030mg		2 to 10min	40 to 240min	340 to 440min
	45kg or 100lbs	6.5 Carpules	58.5mg:0.059mg				
	68kg or 150lbs	9.8 Carpules	88.4mg:0.088mg				
	90kgs or 200lbs	10 Carpules	90mg:0.09mg				
Articaine 4% with 1:100,000 Epinephrine	23kg or 50lbs	2.2 Carpules	161mg:0.04mg		8 to 10min	60 to 90min	190 to 230min
	45kg or 100lbs	4.4 Carpules	315mg:0.079mg				
	68kg or 150lbs	6.6 Carpules	476mg:0.119mg				
	90kgs or 200lbs	6.9 Carpules	500mg:0.125mg				

Hygiene Materials

There are many different products that can be used to help manage a prophylaxis or SRP case. These materials help with educating the patient about their teeth, along with cleaning, and protecting them.

1. **Disclosing Agents and Prophylaxis Pastes**

The goal of a prophylaxis paste is to gently polish and clear the teeth of debris removed from proper scaling.

---

a. ***GUM Red Cote Tablets*** *(Butler)*

- Visualizes dental plaque buildup to show where more brushing and flossing are needed
- Cherry flavored



Instructions for Use

1. Chew one tablet, swish for 30 seconds. Do NOT swallow.
2. Spit and rinse
3. Remaining Red areas indicate harmful plaque
4. Educate the patient on brushing and proximal plaque removal
5. Thoroughly remove the dark red color by brushing and flossing

---

b. ***Clinpro Prophy Paste*** *(3M ESPE)*

- Medicinal Ingredient: Sodium Fluoride (1.23% Fluoride Ion)
- Unit dose cups contain enough Prophylaxis Paste for a single procedure and should only be used once.
- Comes in Mint and Bubble gum



Instructions for Use

1. Using a prophy angle, fill the cup with the paste while running the angle at low speed
2. Apply paste gently to teeth
3. Throughout procedure, sufficient paste should be maintained on the cup
4. Using steady, slow speed, flare the cup during engine polishing to ensure full coverage
5. Saliva evacuation is recommended to avoid excessive dilution and swallowing of the paste

2. Varnish

Varnish products are used to reduce dental hypersensitivities and can be used as adjuncts to the caries control plan.

---

a. ***Prevident Varnish*** *(Colgate)*

- Medicinal Ingredient: 5% Sodium Fluoride (22,600 ppm fluoride)
- Advertised as hypersensitivity relief
- Contains xylitol
- Comes in Caramel Sundae, Grape Bubble Gum, Raspberry, and Mint
- The material hardens on contact with saliva and will coat the teeth for several hours, which occludes the openings of the dentinal tubules



Instructions for Use

1. Wash and dry teeth
2. Mix well prior to application
3. Apply product with supplied brush
4. Thin excess varnish on the tooth's surface until the varnish surface is dry
5. Instruct patients to only eat soft foods for 2 hours after treatment

---

b. ***CavityShield 5% Sodium Fluoride Varnish*** *(3M ESPE)*

- Medicinal Ingredient: 5% Sodium Fluoride (22,600 ppm fluoride)
- Advertised as treating hypersensitivity teeth, exposed dentin, and root surface sensitivity



Instructions for Use

1. Wash and dry teeth
2. Mix well prior to application
3. Apply product with supplied brush
4. Thin excess varnish on the tooth's surface until the varnish surface is dry
5. Instruct patients to only eat soft foods for 2 hours after treatment

---

c. ***Advantage Arrest*** *(Elevate)*

- 38% Silver Diamine Fluoride
- Unit dosing available
- Used for the treatment of dentinal hypersensitivity on a localized basis.
- **Use caution**: This material yielded positive cytotoxicity in standard testing. It should not get anywhere other than the intended treatment site! In addition to this, it will also stain skin and clothes!



Instructions for Use

1. Isolate the affected area.
2. Clean and dry the affected tooth surface.
3. Dispense 1-2 drops of solution into a dappen dish.
4. Use a microbrush or applicator to brush the tooth surface with the material.
5. Air dry.
6. One or two reapplications may be administered at intervals of one week.

3. Miscellaneous

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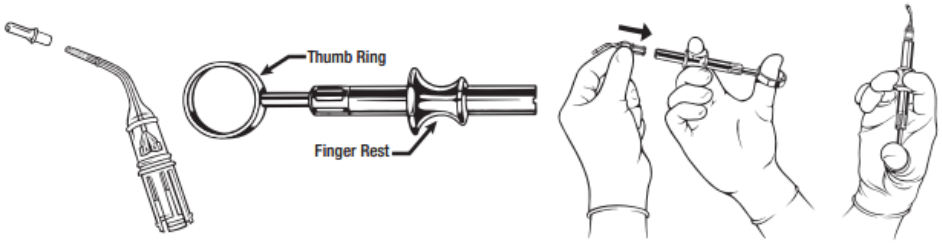
a. ***Arestin*** *(Orapharma)*

- Active Ingredient: Minocycline HCl
- Each unit dose delivers equivalent of 1 mg of minocycline free base
- Indicated as an adjunct to scaling and root planning
- Contraindicated in patients with sensitivity to Minocycline or Tetracycline
  - Tetracycline drugs should NOT be used during tooth development! Therefore, not to be used in children and pregnant or nursing women
- Administration does NOT require anesthetic, but it may be considered depending on patient's tolerance



Instructions for Use

1. Remove disposable cartridge from pouch and connect to the handle mechanism
2. Insert the unit-dose cartridge to the base of the periodontal pocket and press the thumb ring to expel the powder while gradually withdrawing the tip
3. The hand mechanism should be sterilized; however, the Arestin will not need to be removed as it is bioresorbable
4. No dressing or adhesive is required, but instruct patients to avoid chewing hard, crunchy, or sticky foods with the treated teeth for 1 week. Interproximal cleaning of the area should also be postponed for 10 days



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b. ***GUM Paroex CHX Rinse Al-Free*** *(Henry Schein)*

- Active Ingredient: 0.12% Chlorhexidine Gluconate
- Indicated to help treat gingivitis
- Proper studies haven't been done on effects of Paroex on pregnant or nursing mothers. Clinical effectiveness and safety haven't been established for children under 18 as well. Take caution when prescribing to these patients
- Most common adverse effects are staining of teeth, increased calculus formation, and alteration in taste perception.
- Usually prescribed following a dental prophylaxis in a 16 fl oz bottle



Instructions for Use

1. Dispense ½ fl oz of undiluted Paroex
2. Rinse for 30 seconds after brushing
3. Expectorate.
4. Instruct patient *not* to eat immediately after rinsing

Impression Materials

Despite the dental world becoming more and more digital, clinicians still heavily rely on various impression materials for everyday use. From diagnostic casts to definitive casts, these materials will be used to capture the oral cavity. Students use these materials to learn about and create crowns, bridges, RPDs, and Dentures.

1. Tray Adhesives

Whether taking diagnostic or definitive impressions, tray adhesive is often used to ensure the material stays in the tray without distortion.

---

a. ***Tac-Tray Adhesive Liquid*** *(Bosworth)*

- This material is a tray adhesive used in conjunction with alginate impressions.



Instructions for Use

1. Paint or spray adhesive over all surfaces that will directly contact the alginate
2. Wait until adhesive becomes tacky
3. Continue with impression

---

b. ***VPS – Tray Adhesive*** *(Kerr)*

- Rubber based tray adhesive for PVS impressions



Instructions for Use

1. Stir well and paint adhesive over all surfaces that will directly contact the impression material
2. Allow adhesive to air dry for 10-20 minutes until tacky and dull
3. Continue with impression

---

c. ***AccuDent XD Tray Adhesive*** *(Ivoclar Vivadent)*



Instructions for Use

1. Stir well and paint adhesive over all surfaces that will directly contact the impression material
2. Allow adhesive to air dry for 3-5 minutes
3. Continue with impression

2. Diagnostic Impressions

Diagnostic impressions are taken on almost every patient at Western U to help students treatment plan their cases. The casts formed from these impressions provides a way for students to evaluate a case and present their findings without the patient present. They will be used to construct pull-down matrices and custom trays, along with providing a prop to explain treatment to their patients.

---

a. ***Identic Alginate*** *(Kerr)*

- Dust Free, unit dosed alginate
- Premeasured packets of alginate for a single impression. The water measure is incorporated as part of the package and eliminates the canister, scoop, and water vial
- At WesternU, the unit dosing material will be used primarily in the clinical setting when working with a patient



Instructions for Use

1. Fluff the alginate by shaking the packet back and forth
2. Cut the unit dose packaging on the horizontal dotted line
3. Dispense the powder portion into a mixing bowl
4. Fill the other portion to the line with water
5. Mix thoroughly for 30 seconds
6. Fill the impression tray and quickly seat
7. Remove the impression after 3 min and 30 sec

	Mixing Time	Working Time	Setting Time
Regular Set	30 sec	2:20 min	3:30 min

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b. ***HSI Dust Free Alginate*** *(Henry Schein)*

- Irreversible hydrocolloid material
- At WesternU, the bulk alginate material will be used primarily in the laboratory setting without patients



Instructions for Use

1. Fluff the alginate
2. Fill the supplied scoop (8g) and level with a spatula
3. For each scoop of powder, prepare 1/3 measure of water (19mL)
4. Mix thoroughly for 45 seconds
5. Fill the impression tray and quickly seat
6. Remove the impression 1 minute after it is no longer sticky
7. Rinse, clean, and pour immediately, or wrap in a damp towel



	Mixing Time	Working Time	Initial Setting Time	Setting Time
Regular Set	45 sec	2:00 min	2:15 min	3:30 min

c. ***AlgiNot*** *(Kerr)*

- A-silicone impression material intended as an alternative to traditional alginate materials
- Suitable for diagnostic impressions
- Work time: 45sec from start of mixing
- Removal time: 2min 30sec from start of mixing



Instructions for Use

1. Prepare patient and tray for an impression. VPS adhesive should be used with impression tray
2. Insert cartridge into dispensing gun and bleed before attaching the mixing tip
3. Attach appropriate mixing tip
4. Squeeze trigger with moderate and even pressure to begin mixing material and dispense into the desired tray
5. Once set, remove tray with a snap and disinfect tray

3. Definitive Impressions

Once a treatment plan is in place and definitive treatment has begun, definitive impressions will often follow in order to capture the precise detail needed for long term prostheses.

---

a. ***AccuDent XD Impression Material*** *(Ivoclar Vivadent)*

- A two viscosity irreversible hydrocolloid impression material
- Two dual-phase alginate materials for a tray and syringe
- Intended for partial and complete denture impressions

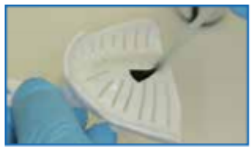


	SYRINGE ALGINATE	TRAY ALGINATE
Color	Orange	Blue (chromatic)
Mixing Ratio (Powder/Water)	9 g Powder/21 ml Water	24 g Powder/44 ml Water

	SYRINGE ALGINATE	TRAY ALGINATE
Mixing Time	45 sec	45 sec
Working Time	2:00 min	1:30 min
Initial Setting Time	3:30 min	2:30 min
Setting Time	3:45 min	2:45 min
*Water Temperature	73°F/23°C	73°F/23°C

Mixing Sequence

1. Apply Tray Adhesive. **Let dry for 3-5 mins.**



2. Mix the Syringe Gel with **21ml of 73°F water.**



3. Spatulate the Syringe Gel for **45 seconds.**



4. Backload the syringe and replace the plunger. Set aside syringe.



5. Mix the Tray Gel with **44ml of 73°F water.**



6. Spatulate for **45 seconds** or until smooth.



Procedure

7. Load the tray from posterior.



8. Syringe Gel onto dentition, posterior to anterior. Repeat for opposing arch.



9. Without tip, dispense Syringe Gel onto vestibules and a small amount into the anterior vault.



10. Place tray front to back. Do not over-seat. **Total Set Time: 3 mins 45 secs** from mixing of Syringe Gel.



11. Remove tray carefully and rinse. Do not pull on handles. **Disinfect and pour model immediately.**



b. ***Aquasil Ultra Monophase*** *(Dentsply)*

- Quadrafunctional hydrophilic addition reaction silicone elastomeric dental impression material
- This material can be used for all dental impression techniques
- It has < 0.50% linear dimensional change, ≥ 98% recovery from deformation, and can reproduce detail to 20 microns



Instructions for Use

1. Assure hemostasis and a clean working field
2. Add tray adhesive to custom tray
3. Insert desired cartridge into gun and bleed before installing the mixing tip.
4. For unit dose cartridges, insert into the syringe barrel, snap the circular disk off, and insert the cartridge into the mixing tip.
5. Dispense material into a tray or directly into the mouth with slow, steady pressure.
6. Insert filled tray into the mouth and press tray gently into position.
7. Allow to set based on the below setting times.
8. Remove tray slowly and disinfect

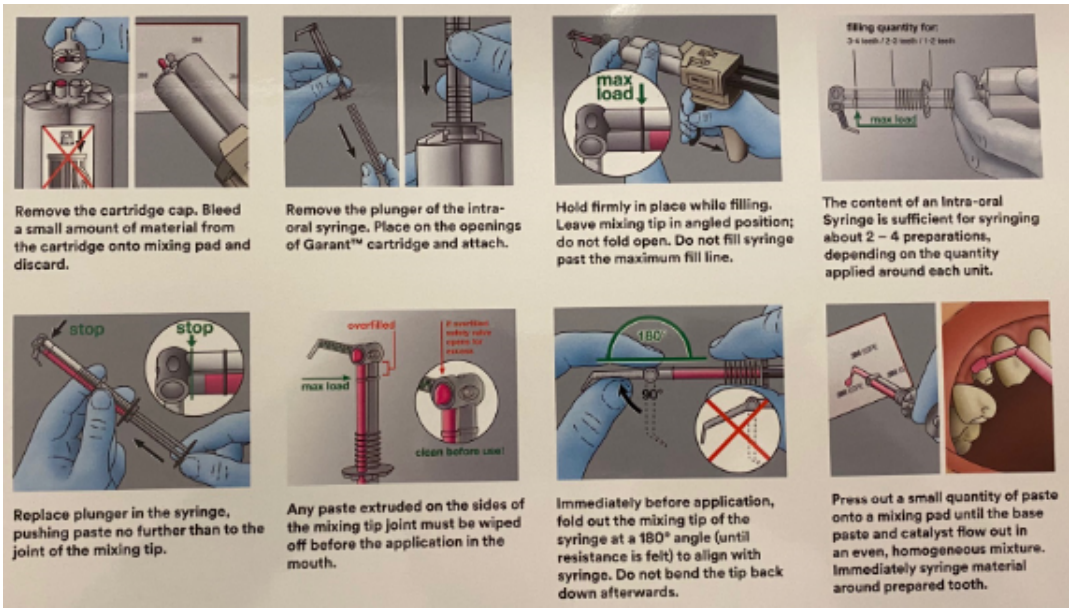
	Working Time at Room Temperature	Working Time Intraorally	Setting Time (From start of mixing)
Regular Set	2min 15sec to 2min 45sec	1min 10sec Max	5min

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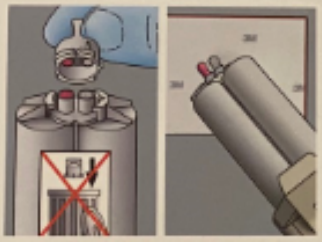
c. ***Imprint*** *(3M ESPE)*

- VPS Impression material used for fixed prosthodontics
- Advertised for improved hydrophilicity

	Color	Working Time at Room Temperature	Working Time Intraorally	Setting Time Intraorally
Imprint 4 Penta Heavy	Blue	2:00 min	--	2:00 min
Imprint 4 Light	Pink	--	1:00 min	2:00 min



Steps for using Imprint™ 4 VPS Impression Material with the Double-Bite Technique



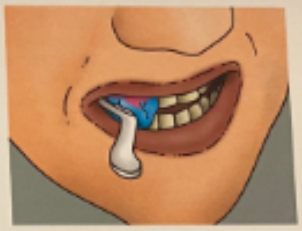
Before adding the mixing tip to the cartridge, extrude a small amount to ensure the base and catalyst are evenly extruded, and discard.



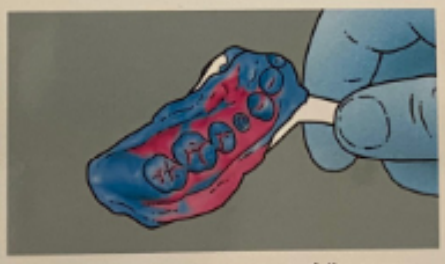
Dispense heavy-body material into the tray, keeping the mixing tip immersed in the paste to avoid formation of air bubbles.



Syringe wash material around the margin.



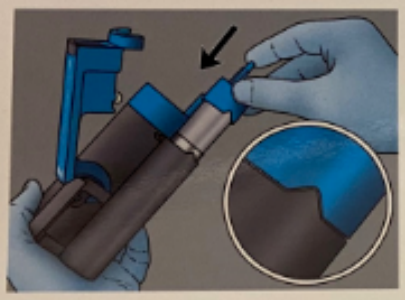
Place tray and have patient gently bite down, and hold. Follow recommended setting time in the mouth using a timer.



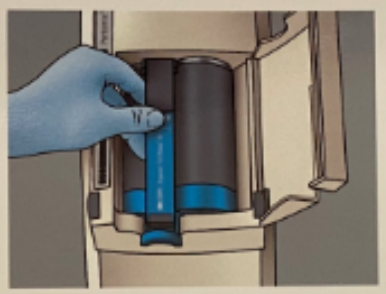
When material is completely set, carefully remove tray from mouth and evaluate the final impression.

3M

Steps for using Imprint™ 4 VPS Impression Material in a 1-Step Impression Technique



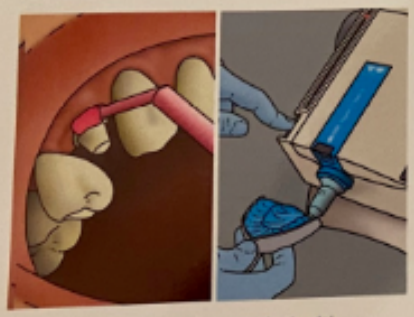
Place the Penta™ foil bags into the designated Penta™ cartridge.



Insert the loaded cartridge into the Pentamix™ Automatic Mixing Unit.



When using new foil bags, start mixing and extrude a small amount until paste is homogeneous in color. Discard.



While the tray is being loaded with heavy-body or putty impression material, apply wash material around the preparation using a wiggle and stir motion.



Insert the loaded tray straight and evenly into mouth—avoid rotation. Don't let patient bite down on tray. Immobilize the tray using passive pressure. Follow recommended setting time in the mouth using a timer.



When material is completely set, carefully remove tray from mouth and evaluate the final impression.

4. Bite Registration

Bite registration materials allow students and dentists to properly articulate their casts. It becomes especially important when a patient has a questionable occlusion or multiple missing teeth.

---

a. ***Blu-Bite HP Regular Set*** *(Henry Schein)*

- Rigid bite registration PVS
- This material will be the "go-to" bite registration material for the Sim Lab and Clinic



Instructions for Use

1. Insert the cartridge into the dispensing gun and apply the tip.
2. Bleed to ensure proper mixing
3. Dispense onto the mandibular occlusal surfaces of the teeth and move the mandible into central occlusion
4. Material will set in ≥ 40 seconds
5. Ensure material is set, remove, and disinfect

---

b. ***ACU-flow*** *(Great Lakes Orthodontics)*

- Syringeable VPS material
- Working time: 15sec from start of mixing
- Total cure time: 1min from start of mixing
- This material will be used with the cases involving the prosthodontic faculty



Instructions for Use

1. Insert the cartridge into the dispensing gun and bleed the cartridge
2. Attach the auto-mix tip
3. Dispense where desired on the teeth and move the mandible into central occlusion
4. Ensure material is set, remove, and disinfect. Exchange auto-mix tip for the cartridge cap

---

c. ***Superbite*** *(Bosworth)*

- Zinc Oxide Eugenol paste
- Blue, two-component material
- Working time: 45sec from start of mixing
- Total cure time: <3min from start of mixing
- This material is only available for lab work with the prosthodontic faculty



Instructions for Use

1. Extrude equal volumes of catalyst and base onto a mixing pad
2. Mix quickly to uniform color (45 seconds)
3. Apply to upper and lower segments of gauze tray
4. Insert and place on occlusal surfaces of lower teeth
5. Have patient gently close into a comfortable, natural bite. Material will set in less than 2 min from placement
6. Rinse with cold water to remove saliva and disinfect

Restorative Materials

Dentistry thrives on restorative materials. From amalgam, to glass-ionomers, and resins, restorative materials are ubiquitous in the dental world.

1. Hemostatic Agents

During many restorative procedures, the dentist will need to combat heme. Heme leads to decreased visual fields, decreased bond strengths, and can ruin impressions.

---

a. ***Hemodent*** *(Premier Dental)*

- Buffered aluminum chloride, epinephrine free hemostatic product



Instructions for Use

1. Apply Hemodent to cotton pellet, microbrush, or pre-saturate a retraction cord
2. Place in area of needed heme control
    - If this is a cotton pellet, apply pressure to area and let sit until a clot has formed before removing and rinsing
    - If this is a retraction cord, place in the sulcus of the tooth and proceed
3. Rinse area thoroughly after use

---

b. ***Viscostat*** *(Ultradent)*

- 20% Ferric Sulfate equivalent hemostatic solution
- Viscostat will create a visible coagulum



Instructions for Use

1. Verify flow before applying intraorally
2. Express and scrub material into area of needed hemostasis
3. As coagulum forms, gently rinse away with air/water spray.
4. Hemostasis is achieved when no new coagulum forms
5. Recommended contact time is 1 to 3 minutes with a maximum of 10 to 20 minutes
6. Thoroughly rinse once finished.

2. Temporary Restoration Materials

Temporary restorations are a common occurrence in dentistry due to the unpredictable nature of emergencies. A teaching environment has an added use for these materials as students may not be able to finish their procedures in the designated time slot. In these instances, temporary restorative materials are used.

---

a. ***Fermit N*** *(Ivoclar Vivadent)*

- Light-curing single-component material designed for temporary restoration of Class I and II restorations
- Does NOT require a cement



Indications

1. Deep inlay preparations
2. Relining material for temporary crowns/bridges
3. Temporarily sealing implant screw access openings
4. Large preparations (onlays)

Contraindications

1. Unsuitable as temporary crown and bridge material
2. Should not be in the mouth for more than 1 month
3. Patients sensitive to the material or ingredients

Instructions for Use

1. Prepare tooth
2. Dry and place matrix if indicated
3. Clean with water
4. Apply desensitizer when indicated
5. Dry with air
6. Place Fermit N
7. Increments of up to 4mm can be cured in 10 second intervals
8. Remove matrix
9. Verify/adjust occlusion

---

b. ***Cavit*** *(3M ESPE)*

- Self-curing material for temporary filling of cavities
- Patients should avoid any exposure to chewing pressure for 2 hours after application



Instructions for Use

1. Prepare tooth as needed based on procedure
2. Once ready for provisional restoration, use an instrument to fill the required quantity into the wet cavity
3. Shape as needed. Material will harden after a few minutes

c. ***ZOE B&T*** *(Dentsply Sirona)*

- Polymer-reinforced zinc oxide-eugenol restorative material
- Designed for intermediate restorations or as a base that does *not* contain resin components



Instructions for Use

1. Prepare tooth as needed based on procedure. Cavity preparation must provide the mechanical retention of the product. Place matrix and wedge when indicated and gently dry preparation
2. Once ready for provisional restoration, dispense two drops of liquid onto a mixing pad
3. Tumble powder bottle to assure uniform bulk density and dispense 0.2g of powder per drop of liquid onto the pad. Keep powder and liquid separate until ready to mix
4. Spatulate 50% of powder into all of the liquid quickly and thoroughly. Bring the remaining powder into the mix in 2 or 3 increments and spatulate thoroughly. Mixing should take about 1 minute
    - A proper mix will appear stiff but with a smooth and adaptable working consistency
    - If too stiff/dry, add liquid. If sticky and soft, add powder
5. Carry material to preparation with an appropriate instrument. Instrument may be dipped in dry powder left on the pad to facilitate placement and minimize sticking
6. Shape as needed and smooth with light burnishing. Material will harden after a few minutes. Once set, remove matrix and adjust occlusion

3. Amalgam

- Even with new and exciting materials continually coming to the dental field, amalgam remains a viable option for dental restorations.
- According to the ADA, "Dental amalgam is considered a safe, affordable and durable material that has been used to restore the teeth of more than 100 million Americans."

---

a. ***Dispersalloy*** *(Dentsply)*

- Designed for use in stress bearing restorations (Class I and Class II)
- Particle type: Admixture of lathe-cut rods and silver-copper spheres
- Recommended alloy:mercury is approximately 1:1
- Any moisture may adversely affect the strength and corrosion resistance of the amalgam due to the zinc.
- Composition:

		1 Spill	Weight %
Powder	**Silver**	265mg	69%
	Tin	69mg	18%
	Copper	46mg	12%
	Zinc	4mg	1%
Liquid	**Mercury**	400mg	

- Timing:

	Regular Set	Fast Set
Condense Time	3min 30sec	2min 30 sec
Set Time	4min 30sec	3min



Instructions for Use

1. Prepare tooth under rubber dam isolation. Rinse and dry
2. If a liner, base, or bonding is needed, place according to manufacturer's instructions
3. Triturate the amalgam. Place capsule into the amalgamator and set time according to amalgamator specifications
   a. If excessively splashy/wet, INCREASE time
   b. If excessively dry, DECREASE time
4. Twist and open capsule. Drop mass into an amalgam well and begin placement and condensation
5. Condense, burnish, and carve
6. Polishing should be done 24 to 48 hours after setting

4. Glass Ionomers and Resin Modified Glass Ionomers

- Glass Ionomers revolutionized the dental world. Introduced by Wilson and Kent in 1972, the combination of a silica and alumina powder with a polyacrylic acid led to a material able to chemically bond to a tooth. The acid/base reaction continues to harden over time and allows for fluoride release. This promising material was biocompatible and cariostatic.
- With the advent of resins, glass ionomers began undergoing a major shift. The downsides of the glass ionomer were its early solubility and weaker overall bond strength to the tooth. Resins showed improvements in both of these areas.
- Many began combining resins with glass ionomers, forming resin modified glass ionomers (RMGIs) and compomers.
- There is not a strict definition clarifying the difference between these materials; however, in general an RMGI is a glass ionomer with resin infiltration, while a compomer is usually described as a resin with glass ionomer infiltration. The difference is subtle, but again, there are no strict definitions for either.

---

a. ***GC Cavity Conditioner*** *(GC America)*

- 20% Polyacrylic Acid for Dentin Pretreatment
- This is the conditioner that will be used to prepare the tooth for Fuji products
- It works by removing the smear layer of the tooth



Instructions for Use

1. Apply conditioner to Smear Layer of tooth.
2. Allow to react for 10 seconds
3. Rinse thoroughly with water
4. Blow dry for two or three short intervals or dab dry with a cotton pellet. Do NOT desiccate/over dry
   - Over drying can lead to post-op sensitivity!

---

b. ***Equia Forte Fil*** *(GC America)*

- Radiopaque posterior glass ionomer restorative cement in a capsule (GI)
- Despite the manufacturer stating this material can be used as a core build-up, research into the material clearly shows it should be avoided for this purpose



Indications

1. Class I and II restorations on deciduous teeth
2. Intermediate restorations for heavy stress Class I and II restorations (Sandwich Technique)
3. Class V and root surface restorations

Contraindications

1. Pulp capping
2. Sensitivity to the material (Contact Physician)

c. ***Fuji II LC*** *(GC America)*

- Radiopaque light cured (LC) reinforced glass ionomer restorative cement in a capsule (RMGI)
- Despite the manufacturer stating this material can be used as a core build-up, research into the material clearly shows it should be avoided for this purpose



Indications

1. Class III and V restorations, particularly cervical erosion and root surface caries
2. Primary teeth
3. Geriatric applications
4. As a base or liner

Contraindications

1. Pulp capping
2. Sensitivity to the material (Contact Physician)

---

d. ***Equia Forte Coat*** *(GC America)*

- Light cured self-adhesive, wear resistant coating
- Used to prevent water and saliva contamination to exposed surfaces of glass ionomer cements during the first 24 hours after placement.

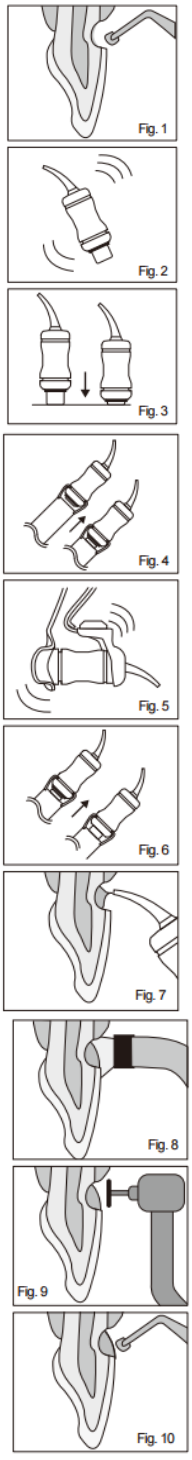


Instructions for Use

1. Place Glass Ionomer according to manufacturer's instructions
2. After shaping is completed, roughen the surface with a fine diamond.
3. Clean and dry the tooth. Dispense and brush Equia Forte Coat over all exposed glass ionomer and adjacent surfaces
    - If uncut enamel will be covered, it should be etched with UniEtch.
    - Material should be brushed on immediately (within 1 minute)!
4. Light Cure for 20 seconds.

Chart 1: Placement of Fuji II LC and Equia Forte Fil

Fuji II LC



Fuji II LC and Equia Forte Fil Instructions for Use

1. **Prepare tooth**
   - Wash and dry
   - Do NOT desiccate
   - *Cavity Conditioner* to remove smear layer
2. **Activate/Mix capsule**
   - Shake/tap capsule
   - Push plunger all the way in
   - *Fuji II LC ONLY:* Place in Capsule Applier and click once
   - *Both:* Remove and mix in triturator for 10sec (4,000 RPM)
3. **Restorative Technique**
   - Load into Capsule Applier
   - Click twice
   - Remove surface moisture (Do NOT desiccate!)
   - Extrude into preparation
   - Contour
   - *Fuji II*, Light cure for 20 seconds
   - *Equia Forte Fil has a 1:15 min* working time
4. **Finish**
   - Finish and polish
   - Apply *Equia Forte Coat*
   - Light cure for 20 seconds

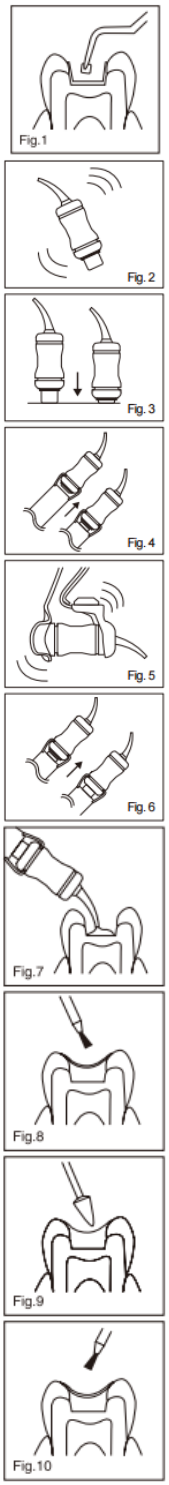
Fuji II LC

DIRECTIONS FOR USE	
Powder / Liquid Ratio (g/g)	0.33 / 0.10
Mixing Time (sec.)	10"
Working Time (min., sec.)	3'15"
Light Curing Time (sec.)	20"
Depth of Cure (A2) (mm)	1.8

Equia Forte Fil

DIRECTIONS FOR USE	
Powder / Liquid Ratio (g/g)	0.40 / 0.13
Mixing Time (sec.)	10"
Time of initial extrusion after mixing	10 seconds maximum
Working Time (min., sec.)*	1' 15"
Net Setting Time (min., sec.)*	2' 00"
Final Finishing Commencing Time (37°C, after starting mix)(min., sec.)	2' 30"
VLC Curing Time of EQUIA Forte Coat	20"

Equia Forte Fil



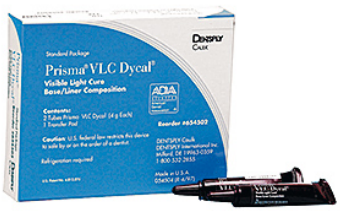
5. Liners

- For the deep, carious lesion, liners are often a useful material to help with post-op sensitivity and to promote healing.
- Definitions:
  - **Liners**: A thin layer of material (less than 0.5mm), placed on dentin either for its therapeutic effect, or to create a physical barrier, or both.
  - **Bases**: A replacement material for missing dentinal tooth structure and is typically greater than 0.5mm in thickness.
    - The term "Base" is more of a historical term when dentin needed to be replaced before additional materials could be added. With the advent of new materials, this term is much less common and will not be utilized in this manual.

---

a. ***Prisma VLC Dycal*** *(Dentsply)*

- Calcium Hydroxide Liner
- Rigid-setting, self-curing material for direct and indirect pulp capping
- Does NOT inhibit polymerization of acrylic and composite materials
- Despite manufacturer instructions for indirect pulp capping, this material will be used much more commonly for direct pulp capping



Direct Pulp Capping Technique

1. Prepare tooth under rubber dam isolation, removing all caries
2. Rinse with 2.6-5% NaOCl. Control bleeding with cotton pellet moistened with saline
3. Dry with a cotton pellet. Do NOT desiccate
4. Dispense onto a mixing pad.
5. Place mix directly on the exposed pulp and cavity dentin. This layer should be 0.8-1.0 mm thick.
6. Light cure for 20 seconds.
7. Remove any excess material.
8. If indicated, seal Dycal with a layer of Vitrebond.
9. Continue with provisional or permanent restoration

Indirect Pulp Capping Technique

1. Prepare tooth under rubber dam isolation, removing all caries
2. Dry with a cotton pellet. Do NOT desiccate.
3. Dispense onto a mixing pad.
4. Place mix into the deepest area of the cavity dentin. This layer should be 0.8-1.0 mm thick.
5. Light cure for 20 seconds.
6. Remove any excess material.
7. Continue with provisional or permanent restoration

---

b. ***Lime-Lite Enhanced*** *(Pulpdent)*

- A moisture tolerant, light cure, radiopaque cavity liner and base with self-adhesive properties. It releases calcium, phosphate, and fluoride ions
- This product is NOT indicated for direct pulp capping!



Instructions for Use

1. Prepare cavity, rinse, and dry, but do not dessicate
2. Attach a pink 20-gauge dispenser tip
3. Apply 1-1.5 mm Lime-Lite Enhanced to the dentin
4. Using a cotton pellet or microbrush, saturate with product
5. Light cure for 20 seconds
6. Continue with restoration. This material can be etched and is compatible with composites

c. ***Vitrebond Plus*** *(3M ESPE)*

- Resin Modified Glass Ionomer
- Indicated as a lining under composite, amalgam, ceramic, and metal restorations.
- This product is NOT indicated for direct pulp capping!



Instructions for Use

1. Prepare tooth, removing all decay under rubber dam isolation
2. Rinse and dry. Do NOT desiccate!
3. Do NOT pretreat the dentin, or risk decreased adhesion!
4. Dispense 1 level powder scoop onto a mixing pad
5. Hold liquid vial vertically and dispense 1 drop
6. Mix until there's a smooth consistency and glossy appearance. (10-15 seconds)
7. Apply using a ball applicator to dentin surface. This layer should be 0.5 mm thick or less Vitrebond has a 2 minute and 30 second working time
8. Cure for 20 seconds
9. Continue with bonding steps, starting with etching

---

d. ***MicroPrime G*** *(Danville)*

- HEMA and glutaraldehyde desensitizing agent
- Helps kill bacteria, alter nerve responses, and aids bonding primers in penetrating etched dentin



Non-Bonded Restorations

1. Clean tooth preparation
2. Dry with air
3. Apply MicroPrime G using a brush or cotton pellet. AVOID Soft Tissue!
4. Wait 30 seconds, then dry with air
5. Place restorative material

Bonded Restorations

1. Clean tooth preparation
2. Etch with Ultra-Etch for 15 seconds.
3. Rinse and dry
4. Apply MicroPrime G using a brush or cotton pellet. AVOID Soft Tissue!
5. Wait 30 seconds, then dry or leave moist depending on bonding instructions
6. Proceed with bonding per manufacturer's instructions

---

e. ***Super Seal*** *(Phoenix Dental)*

- Water based potassium oxalate that removes the smear layer, seals the tubules, and desensitizes the tooth
- This product will not harm soft tissue



Instructions for Use

1. Dispense 1 to 2 drops of Super Seal in a dappen dish
2. Using a cotton pellet or microbrush, saturate with product
3. Gently rub desired area for at least 5sec. Do NOT dry!
4. Continue with restoration

6. Adhesive Systems

- Dental adhesives have seen 7 generations of products for bonding our restorative materials to teeth. Each generation has its advantages and drawbacks. While manufacturers find a way to incorporate an entire system into one bottle, many systems that keep each element separate still reign supreme.
- Three basic steps are involved with dental adhesives: Etch, Prime, and Bond
- As the generations progressed, the systems were no longer referred to by these steps, but rather by the number of steps involved.

Etch	Prime	Bond
• Removing the smear layer • Exposing intertubular and peritubular collagen • Opens the tubules in a funnel configuration • Decreases the surface free energy	• Includes both hydrophilic and hydrophobic molecules simultaneously • Envelops the external surface of collagen fibrils • Re-establish surface-free energy to levels compatible with a more hydrophobic restorative material	• Includes monomers that are mostly hydrophobic • Copolymerize with the primer • Penetrates and polymerizes into the interfibrillar spaces to serve as a structural backbone to the hybrid layer

---

a. ***Ultra-Etch*** *(Ultradent)*

- 35% Phosphoric Acid Gel



Instructions for Use

1. Prepare tooth under rubber dam isolation and rinse well
2. Bleed syringe to ensure proper etchant flow prior to intraoral application
3. Extrude etchant onto the surface to be etched and leave for 15 seconds
4. Thoroughly rinse for 15 seconds. Suction first to avoid etchant going to unwanted areas
5. Dry with clean, oil-free, air without desiccating the dentin
6. Immediately proceed with bonding procedure

---

b. ***OptiBond Solo Plus*** *(Kerr)*

- 5th Generation Adhesive (Etch separately, prime and bond in one bottle)
- 15% filled
- Indicated for both direct and indirect restorations



Instructions for Use

1. Prepare tooth (under rubber dam isolation when applicable) and rinse well
2. Shake bottle briefly prior to dispensing
3. Dispense one drop of adhesive into disposable mixing well
4. Etch tooth for 15sec with 37.5% Phosphoric acid etch and rinse thoroughly
5. Dry lightly but do not desiccate
6. Apply OptiBond Solo Plus with applicator for 15sec
7. Air thin for 3sec
8. Light cure for 20sec
9. Continue placing composite

c. ***OptiBond FL Prime*** *(Kerr)*

- 4th Generation Adhesive (Etch, Prime, and Bond are all separate)
- OptiBond FL Primer is to be used with Ultra-Etch, and OptiBond FL Adhesive
- Available in unidose



Instructions for Use

1. Prepare tooth under rubber dam isolation. Rinse and dry well
2. Properly etch preparation using Ultra-Etch
3. Dispense OptiBond FL Primer into a mixing well. If using the unidose, use provided applicator into the yellow unidose capsule
4. Apply material over enamel and dentin with a light scrubbing motion for 15 seconds
5. Air dry for 5 seconds. Surface should be slightly shiny
6. Proceed with bonding and restoring resin restoration

---

d. ***OptiBond Solo Adhesive*** *(Kerr)*

- This material is to be used with Ultra-Etch and OptiBond FL Primer
- Available in unidose



Instructions for Use

1. Prepare tooth under rubber dam isolation. Rinse and dry well.
2. Properly etch using UniEtch
3. Properly prime using OptiBond FL Primer
4. Dispense and apply OptiBond FL Adhesive. If using the unidose, use provided applicator into the black unidose capsule
5. Use air to thin the material and light cure for 20 seconds
6. Continue placing resin



Table 3: Condensed Guide to Adhesive Systems

2-Step	Etch	Prime	Bond		Restoration Used For
Kerr 	**Ultra-Etch**	**OptiBond Solo Plus**			
	1. 15 sec Etch 2. Rinse 5 sec 3. Dry Lightly (do not desiccate)	1. Apply and let sit for 15sec 2. Evaporate with air for 3sec 3. Light cure 20sec 4. Continue with resin			**Resin**
3-Step					
Kerr 	Ultra-Etch	OptiBond FL Primer	OptiBond FL Bond		
	1. 15 sec Etch 2. Rinse 5 sec	1. Scrub surface with material for 15sec 2. Air dry for 5sec	1. Dispense and apply 2. Air thin 3. Light cure 20sec 4. Continue with resin		**Resin**

7. Composite

- After glass ionomers, composites were formulated that revolutionized dentistry – and haven't left. Today, they are still the most esthetic direct restorations available with some of the highest bond strengths in dentistry.

---

a. ***Revolution Formula 2*** *(Kerr)*

- 60% Filled, flowable, light-cure, hybrid resin restorative
- Shades available: A1, A2, A3



Instructions for Use

1. Select shade and prepare tooth (under rubber dam isolation when indicated)
2. Rinse, dry, and prepare matrix and liner as needed
3. Apply adhesive system according to manufacturer's instructions
4. Dispense Revolution Formula 2 with slow, steady pressure, beginning with the deepest area first.
    a. Avoid lifting the tip to minimize air entrapment
    b. Do not leave too long under intensive light or the material will begin to cure!
    c. Fill up to 2mm
    d. When finished, wipe tip against cavity wall as the tip is removed
5. Light cure each increment for 20 seconds
6. Proceed with completing the restoration

---

b. ***Natural Elegance Pit and Fissure Sealant*** *(Henry Schein)*

- Designed for sealing pits and fissures on primary and permanent posterior teeth
- Opaque white in shade



Instructions for Use

1. Clean, isolate the teeth, and blow dry
2. Apply etchant and allow to react for 30 seconds
3. Wash for 20 seconds – do not allow saliva to contaminate
4. Repeat steps 2 and 3 as needed until enamel has a dull, white, and frosty appearance once air dried
5. Apply a syringe tip to Natural Elegance Pit and Fissure Sealant and express into desired pits and fissures. A microbrush may be used to agitate material and remove excess as needed
6. Light cure for 30 seconds
7. Wipe with a cotton roll to remove the oxygen inhibited layer
8. Check margins with an explorer and adjust occlusion as needed

---

c. ***Embrace WetBond Pit & Fissure Sealant*** *(Pulpdent)*

- Fluoride releasing, light curing, resin-based sealant material that bonds chemically and micromechanically to slightly moist teeth



Instructions for Use

1. Clean, isolate, and etch the teeth for 15 seconds
2. Rinse well. Lightly dry, but leave tooth surface slightly moist
3. Attach applicator tip and apply Embrace sealant to desired locations
4. Light cure for 20 seconds
5. Check margins with an explorer and adjust occlusion as needed

d. ***Herculite Ultra*** *(Kerr)*

- Universal nanohybrid dental composite
- Shades available:
  - Enamel: A1, A2, A3, A3.5, A4, B1, B2, B3, B4, C1, C2, C3, C4, D2, D4
  - Dentin: A2d, A3d, B1d, C2d
  - Incisal Shade



Instructions for Use

1. Select shade and prepare tooth (under rubber dam isolation when indicated)
2. Rinse, dry, and prepare matrix and liner as needed
3. Apply adhesive system according to manufacturer's instructions
4. Place liner material or flowable resin when indicated
5. Dispense Herculite Ultra either onto an instrument or directly into the preparation with gentle, even pressure, beginning with the deepest area first
   a. Avoid lifting the tip to minimize air entrapment
   b. Do not leave too long under intensive light or the material will begin to cure
   c. Fill in 2mm increments
   d. When finished, wipe tip against cavity wall as the tip is removed
6. Light cure each increment for 20 seconds
7. Once filled, proceed with finishing and polishing the restoration

---

e. ***Clearfil Photo Core PLT*** *(Kuraray)*

- Light-cure, radiopaque core build-up composite material
- This material will be used primarily for core build-ups
- Despite the manufacturer insert describing cure depths of 4+mm, this material should follow the 2mm, 20sec cure regimen



Instructions for Use

1. Select shade and prepare tooth (under rubber dam isolation when indicated)
2. Rinse, dry, and prepare matrix and liner as needed
3. Apply adhesive system according to manufacturer's instructions
4. Place liner material or flowable resin when indicated
5. Dispense Clearfil Photo Core either onto an instrument or directly into the preparation with gentle, even pressure, beginning with the deepest area first
   a. Avoid lifting the tip to minimize air entrapment
   b. Do not leave too long under intensive light or the material will begin to cure
   c. Fill in 2mm increments
   d. When finished, wipe tip against cavity wall as the tip is removed
6. Light cure each increment for 20 seconds
8. Once filled, proceed with finishing and polishing the restoration

8. Finishing Systems

- Once a restoration has been placed, rarely will it not need to be modified. After gross reduction using finishing carbides and finishing diamonds has been completed, it's time to refine the restoration using finishing discs, strips, and polishers.

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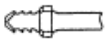
a. ***Super-Snap Polishing Discs*** *(Shofu)*

- These discs are meant for contouring and finishing composites
- They use silicon carbide and aluminum oxide to create 4 different grits
- These come in 8mm disc size in all 4 grits



Instructions for Use

1. Prepare tooth under rubber dam isolation.
2. Place restoration per standard techniques.
3. Perform gross reduction using finishing burs.
4. Once ready for contouring and finishing, insert the Super-Snap Mandrel into the Slow Speed Handpiece and attach the desired Super-Snap disc grit. Base the grit on how much contouring is needed. For more contouring, begin with the coarse grit.
5. Begin to run the handpiece at a slow speed with light pressure.
6. Carefully contour and finish the restoration, working through the different grits. Do NOT skip grits!
7. Once contouring and finishing are complete, move to polishing and completing the restoration.

	Contouring	Finishing	Polishing	Superpolishing
Super-Snap® Disks (50 pcs./box)	 L506 L507 **Black** (coarse)	 L508 L509 L522 **Violet** (medium)	 L501 L519 **Green** (fine)	 L502 L521 **Red** (Superfine)
Super-Snap® Mandrel (6 pcs./box)	 HP - PN 0437	 FG - PN 0438	 CA - PN 0439	

b. ***Maestro Diamond Finishing and Polishing Strips*** *(Henry Schein)*

and

c. ***Diamond Strip Perforated*** *(Axis)*

- Finishing strips that come in various sizes and grits
- Because of the metal design, these strips tend to be more aggressive and have greater risk of tissue laceration. **Be cautious to properly adapt the strip to the contours of the tooth and avoid undue tissue damage**



Instructions for Use

1. Prepare tooth under rubber dam isolation
2. Place matrix and wedges as needed to restore tooth, and begin restoring
3. Once resin has been placed, cured, and the matrix removed, determine the grit of diamond strip required for the restoration based on how much excess remains. More excess means a more aggressive grit should be chosen
4. Carefully slide the strip between the two teeth with the grit facing the restoration to be polished
5. While maintaining the desired contours of the tooth, gently draw the strip back and forth across the grit.
6. If an aggressive grit was chosen to start with, it is necessary to work through the finer grits until the restoration is as smooth as possible. Do NOT skip grits!
7. Interproximal polishing is complete when the margin of the restoration matches the tooth without flash and cannot be detected by the tine of the explorer

---

d. ***Moore-Flex Finishing Strips*** *(E C Moore)*

- These finishing strips come in various sizes and grits as well.
- The grit is an aluminum-oxide coating. These strips will be more flexible and slightly more forgiving than the **Diamond Strips**; however, they will not reduce as aggressively as them either.



- There is no need to use only one system. If the situation calls for both strips to be used, there are no difficulties with this.
- The placement and use of these is the same as the **Diamond Strips** listed above. Please refer there for proper **Instructions for Use**; however, without the serrated saw portion, these strips will not be used for cutting through an interproximal contact

Fixed Prosthodontic Materials

After caries control has been effectively treated, definitive restorations can be placed. The world of crowns and bridges comes with a whole new set of materials to provisionalize and restore teeth.

1. Provisionalization

- After a crown or bridge has been prepared, the patient will require a temporary prosthesis while the permanent one is being fabricated. The following materials are used to provisionalize a tooth for a short time (traditionally less than a month).

---

a. ***Protemp Plus Temporization Material*** *(3M ESPE)*

- Bis-acrylic composite
- Comes in shades A1, A2, A3, A3.5, B3, and Bleach
- 5 minute set time



Instructions for Use

1. Bleed a small amount of material through the mix tip before dispensing into matrix
2. Dispense material into impression/matrix. Fill from the bottom up
3. Seat impression/matrix in mouth and hold firmly in place
4. Remove from mouth between 1:40 and 2:50 min:sec from the onset of mixing
5. Leave material in impression/matrix for 5 minutes
6. Remove oxygen inhibited layer with ethanol
7. Finish and polish as needed
8. Cement with temporary cement when ready

---

b. ***TempBond NE – Unidose*** *(Kerr)*

- Zinc Oxide/Non-Eugenol Cement
- Designed for cementing temporary crowns, bridges, inlays, onlays, and splints
- 1min 30sec working time, with a 7min setting time



Instructions for Use

1. After tooth has been prepared, clean the tooth with water and air
2. Cut along the dotted line and dispense contents of pouch onto mixing pad
3. Mix for 30sec
4. Dry the prepared teeth
5. Apply a thin layer of cement to the internal surfaces of the provisional
6. Firmly seat the restoration
7. After material has set, remove excess

---

c. ***Durelon Carboxylate Luting Cement*** *(3M ESPE)*

- Carboxylate luting cement with relatively neutral pH, low film thickness, and fluoride release
- Used for temporarily cementing crowns, bridges, and inlays/onlays
- 2 min 30 sec working time, with a 10 min setting time



Instructions for Use

1. After tooth has been prepared, clean the tooth with water and air
2. Dry restoration with air
3. Dispense one spoonful of powder to 3 drops liquid onto mixing pad
4. Mix powder and liquid until uniform consistency achieved
5. Apply a thin layer of cement to the internal surfaces of the provisional before mixture loses its gloss
6. Seat the restoration without pressure
7. After material has set, remove excess

2. Materials for Provisional Removal and Crown Seating

- Despite the incredible skill of dental labs, there are occasions when a crown or bridge does not fully seat. Luckily, there are materials to help find these areas and adjust them.

---

a. ***Richwil Crown and Bridge Remover*** *(Almore)*

- This product is a small pellet that is advertised to help remove temporary or permanent crowns; however, not all permanent crowns will be dislodged by one of these
- Evaluate the integrity of the opposing arch before using this product



Instructions for Use

1. Hold remover under warm water for 1 to 2 minutes.
2. Remove once pellet compresses slightly with firm force.
3. Tie to floss and place on the occlusal/incisal of the crown to be removed with firm finger pressure.
4. Ask patient to bite and compress the remover.
5. Hold this position as cold water is sprayed over the remover.
6. Instruct patient to open quickly and forcefully.
7. Remove crown and pellet with gauze.

---

b. ***Occlude*** *(Pascal International)*

- An aerosol spray used for articulation marking
- An excellent aid in seating castings and marks any tooth or restorative surface in either a wet or dry field
- Approved for intraoral use



Instructions for Use

1. Shake well and insert extension tip into valve opening
2. Hold tip 1” from surface to be coated
3. Depress valve gently and lightly to mask desired surfaces
4. Place crown/bridge back on die/tooth
5. Remove crown/bridge and note contact points
6. Adjust crown/bridge and repeat
7. Once adjustments are adequate, remove remaining occlude with soap and water
8. Re-etch/Sandblast intaglio of crown as indicated before permanently cementing

---

c. ***Fit Checker Advanced*** *(GC America)*

- VPES Silicone for checking fit
- Can be used to check the fit of crowns, posts, inlays, and/or dentures



Instructions for Use

1. Squeeze equal lengths of base and catalyst from tubes onto the mixing pad and mix for 20sec
2. Seat appliance. Working time is 1 minute from start of mixing; therefore, hold in position for 1 minute until set
3. Remove prosthesis and analyze/adjust any high spots
4. Repeat
5. Once material presents uniformly throughout denture, remove material and clean before returning prosthetic to patient

3. Primers and Cements for Permanent Fixed Indirect Restorations

- Once the final crown or bridge has been received and all the appropriate adjustments have been made, a permanent cement should be used to lute the crown.

---

a. ***Z-Bond*** *(Danville)*

- Single bottle primer for ceramic and metal surfaces
- Primarily intended for zirconia; however, provides improved adhesion on non-precious metals and alumina ceramics
- Used primarily by the orthodontists at WesternU



Instructions for Use

1. Prepare intaglio of the indirect restoration based on material requirements
   - Metal/Zirconia sand blast and clean
   - Lithium Disilicate/Porcelain etch with HF and clean
2. Dispense a drop of the Z-Bond into a dappen dish
3. Apply primer to intaglio using a microbrush
4. Allow to dwell for 30sec
5. Evaporate the solvent under a gentle air stream for 5 to 10sec
6. Continue with cementation process

---

b. ***Monobond Plus*** *(Ivoclar Vivadent)*

- 3 different methacrylates: silane, phosphoric, and sulfide methacrylate
- A universal primer for conditioning all types of ceramic surfaces



Instructions for Use

1. Prepare restoration (Sand blast/HF/etc depending on material)
2. Apply a thin coat of Monobond Plus to the pre-treated surfaces
3. Allow to react for 60 seconds
4. Disperse any remaining excess with a strong stream of air
5. Continue with cementation process

---

c. ***Ultradent Porcelain Etch and Silane*** *(Ultradent)*

- Viscous, buffered 9% Hydrofluoric Acid with Silane product
- Product is used to prepare porcelain, lithium disilicate, lithium silicate, and other silica-based materials for cementation. It is NOT indicated for use with zirconia
- For lithium disilicates (IPS e.max), manufacturer states etching for 20 seconds.



Instructions for Use

1. Etch clean bonding surface with Porcelain Etch for 90 seconds, rinse, and dry
2. Apply phosphoric acid for 5 seconds to remove debris, rinse, and air dry
3. Apply a puddle coat of Silane to inside surface for 60 seconds, dry, and set prosthesis aside. Do NOT rinse

d. ***Ivoclean*** *(Ivoclar Vivadent)*

- Polyethelene glycol and sodium hydroxide containing cleaning solution
- Used to clean a dental prosthetic of saliva-contaminated restoration surfaces



Instructions for Use

1. Try in restoration and adjust as needed until satisfactory.
2. When ready to cement, start by applying Ivoclean to the bonding surface of the restoration.
3. Allow Ivoclean to react for 20 seconds.
4. Rinse thoroughly and dry with air.

---

e. ***RelyX Luting Plus*** *(3M ESPE)*

- Radiopaque, fluoride-releasing, resin-modified glass ionomer luting cement
- Self-curing with an option for tack light curing of excess cement.
- This cement has a 2min 30sec working time at room temperature.



Instructions for Use

1. Remove the temporary crown, clean the tooth, and gently dry with air. Tooth should be glossy in appearance
2. Try-in the permanent crown and assure proper fit before cementing. If internal adjustments are made, clean debris thoroughly
3. Remove cap and attach the automix tip
4. Apply a thin layer to the intaglio of the restoration
5. Seat the restoration with light pressure
   - Self-curing: remove excess after restoration has been in the mouth for 2min
   - Light-curing: light-cure each surface for 5sec and remove excess with a hand instrument
6. Finish restoration and check occlusion after the crown has been set for 6min

---

f. ***RelyX Unicem 2*** *(3M ESPE)*

- Dual-curing, self-adhesive resin cement for manual mixing
- This cement has a 2min 30sec working time at room temperature
- Before beginning cementation, always refer to product IFU for proper treatment of crown material before cementation



Instructions for Use

1. Remove the temporary crown, clean the tooth, and gently dry with air. Tooth should be glossy in appearance
2. Try-in the permanent crown and assure proper fit before cementing. If internal adjustments are made, clean debris thoroughly
3. Remove cap and attach the automix tip
4. Apply a thin layer to the intaglio of the restoration
5. Seat the restoration with light pressure
   - Self-curing: remove excess after restoration has been in the mouth for 3 minutes
   - Light-curing: light-cure each surface for 2 seconds and remove excess with a hand instrument
6. Finish restoration and check occlusion after the crown has been set for 5 minutes

g. ***Variolink Esthetic DC*** *(Ivoclar Vivadent)*

- Dual-curing luting composite system for the adhesive luting of ceramic, and composite restorations



Instructions for Use

1. Remove temporary crown and clean the tooth of temporary cement
2. Try in the restoration. Variolink Try-In may be used to assist in this process. Do NOT check occlusion at this step
3. Once satisfactory, rinse off the restoration
4. Appropriately pre-treat restoration based on material:
   - For IPS Empress, etch with 5% HF for 60 seconds
   - For IPS e.max, etch with 5% HF for 20 seconds
   - For Zirconia, sandblast the bonding surface and clean with ultrasonic if necessary
   - Apply Monobond Plus for 60 seconds and dry with air
5. Pre-treat tooth:
   - Apply Ultra-Etch for 15 - 30 seconds on enamel and 10 - 15 seconds on dentin and wash away for at least 5 seconds
   - Do NOT over dry dentin! Should appear moist
   - Apply adhesive and scrub for 10 seconds. Air dry
6. Bleed a small amount of material and place a new mixing tip onto the Variolink Esthetic DC
7. Apply mix on the inner surface of the restoration
8. Place restoration and remove excess cement as increasing pressure is applied
   - Light curing can be done in 2 second intervals to decrease set time
9. Maintain pressure for several seconds. Thoroughly light cure from all aspects for another 30 seconds
10. Once set, ensure all excess has been removed, check occlusion, and polish as needed

Table 4: Permanent Cement Indications

	Rely X Luting Plus (RMGI)	Rely X Unicem 2 (Dual Cure Resin)	Variolink Esthetic (Dual Cure Resin)
Metallic Materials			
Full Cast Metal (Gold)	✓	✓	✗
PFMs/MCCs	✓	✓	✗
Porcelain Materials			
Lithium Disilicate (CAD/CAM)	✗	✓*	✓
Feldspathic	✗	✓*	✓
Zirconia	✓	✓*	✓
Veneers	✗	✗	✓
Posts			
Metallic	✓	✓	✗
Fiber-Reinforced	✗	✓	✓
Implants			
Implant Crown	✓	✗	✗

Notes:

- Crowns returned from the lab will often have the intaglio already prepared, while those made through CAD/CAM will need to have the intaglio prepared. If in doubt, discuss with lab technician whether or not preparation is needed.
- ✓*: Rely X Unicem is adequate when retention through preparation is adequate. With minimal mechanical retention, Panavia F 2.0 or Variolink Esthetic is preferred.

4. Posts

- Posts are usually made of metal or fiber that fit into the canal space of a tooth to increase retention and resistance of the restoration/core material**.**
- Therefore, if the core can be retained *without* a post, it should not be used (i.e. more than **50%** of the tooth structure remains).
- Historically, threaded, active posts have been used. As research emerged, threaded posts were found to increase microfractures on the tooth. For those reasons, these types of posts have been dismissed from clinic.
- General measurements:
  - 1mm of radicular dentin around post
  - Minimally 4, but ideally 5mm of adequate apical seal (gutta percha)
  - 2mm circumferential ferrule
  - 2mm for biological width – should NOT be violated!
- This area will be used to discuss the differences between the various posts available, along with each ones' advantages and disadvantages

---

a. ***UniCore*** *(Ultradent)*

- Prefabricated double tapered, translucent glass fiber reinforced composite post system
- Comes in 5 sizes that match the UniCore Drill

Advantages

- One-day placement
- Immediate Provisionalization
- More esthetic

Disadvantages

- Separate core
- Shape may not fit canal well



	Accessory Post	Size 0	Size 1	Size 2	Size 3	Size 4
Apical Ø	0.4 mm	0.6 mm	0.8 mm	1.0 mm	1.2 mm	1.5 mm
Coronal Ø	0.8 mm	1.0 mm	1.15 mm	1.35 mm	1.55 mm	1.75 mm
Taper	0.26°	2.1°	1.8°	1.8°	1.8°	1.3°
Length	16 mm	19 mm	19 mm	19 mm	19 mm	19 mm

Physical properties	UniCore quartz fiber post
Flexural modulus of elasticity (GPa)	43–44
Flexural strength (MPa)	1500–1600
Tensile strength (MPa)	1200
Modulus of elasticity at 30° (GPa)	13 (similar to dentin)
Interlaminate shear strength (MPa)	70–80

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b. ***Parallel-sided Stainless Steel Posts*** *(Henry Schein)*

- Stainless-steel post
- Passive cemented, parallel-sided post with flat head for direct post/core build-ups in one appointment



Advantages

- One-day placement
- Immediate Provisionalization
- Effective with amalgam core

Disadvantages

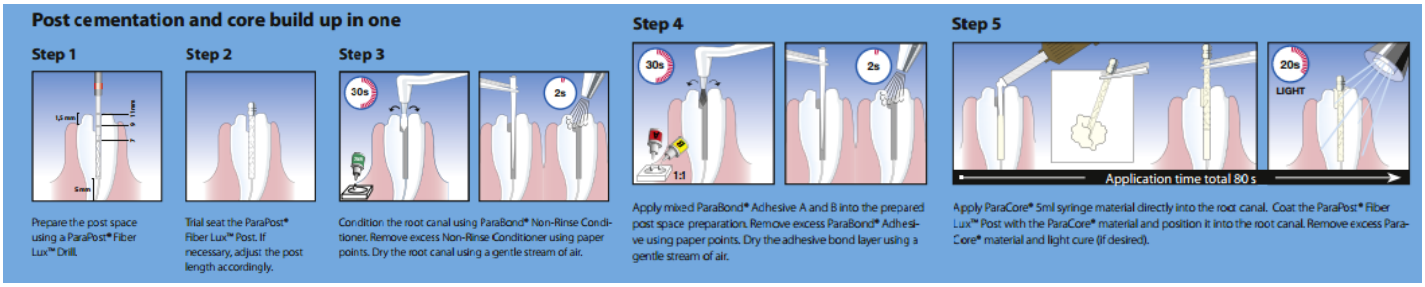
- Separate core
- Shape may not fit canal well

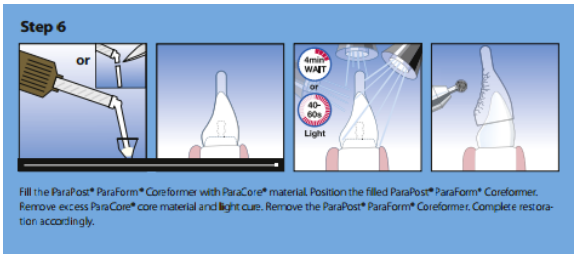
Post Placement

- Proper protocol for placing a prefabricated post. Instructions are almost identical for ***both the fiber post and metal post.***

Standard Procedure for a Prefabricated Post

1. Just like the core build-up and RCT, a rubber dam is mandatory for a successful post placement
2. After RCT, ensure proper ferrule (2.0mm circumferentially). When there is minimal ferrule or multiple missing walls (~50%), a post should be considered.
3. Plan proper diameter and depth of post based on appropriate radiograph
   - At least 1mm of root wall must be maintained
   - At least 4-5mm of gutta-percha must be left apically
4. Remove gutta-percha to preplanned depth using the provided burr set, and/or hot instrument
   - Remove all weakened or unsupported tooth structure
   - Irrigate to clear canal of debris
5. When indicated, create an anti-rotational box
6. Select and place desired prefabricated post. Trim as needed to desired length
7. Clean post with an alcohol wipe. Clean the canal using diluted NaClO, rinse, and dry the canal using Hygienic Paper points
8. Use cement components according to cement instructions for both the canal and post
   - Fiber posts are cemented using either RelyX Unicem 2 or Panavia F 2.0
   - Although metallic posts can be cemented with either an RMGI or a resin cement, the RMGI (Rely X Luting) is preferred
   - A lentulo spiral can be used to help coat the walls of the post space when Rely X Luting is used
9. Seat post and apply pressure for 60 seconds
   - If light curing is necessary, wipe away any excess and cure directly over the post
10. Continue/complete the core and fabricate the final restoration





Removable Prosthodontic Materials

Sometimes teeth need to be replaced, and a fixed option is not viable. Here is where materials for removable prosthodontics will be discussed.

1. **Partial Denture and Complete Denture Adjusting Adjuncts**
   - After a prosthesis has been constructed, it will often require minor adjusting to avoid sore spots. The following products assist in this process.

---

a. ***Mizzy Pressure Indicating Paste (PIP)*** *(Keystone Industries)*

- Placed in removable prostheses to mark high spots



Instructions for Use

1. Dry denture thoroughly
2. Brush PIP over denture in a thin layer so that the brush strokes show
3. Insert denture and place with moderate hand pressure
4. Remove denture and adjust areas that show through the paste
5. Repeat process until high spots no longer mark. Then clean and return appliance to patient

---

b. ***Hydent*** *(Pascal)*

- Placed in removable prostheses to mark high spots



Instructions for Use

1. Dry denture thoroughly
2. Spray over denture in a thin layer
3. Insert denture and place with moderate hand pressure
4. Remove denture and adjust areas that show through the paste
5. Repeat process until high spots no longer mark. Then clean and return appliance to patient

---

c. ***Dr. Thompson Color Transfer Applicator*** *(Great Plains Dental Products)*

- Allows for color transfer from irritated tissue onto removable prostheses



Instructions for Use

1. Wet the colored end of the color transfer applicator
2. Mark the irritated tissue intraorally
3. Insert prosthetic into place with gentle pressure
4. Remove appliance and relieve area indicated on prosthetic

2. Relining Materials

- Occasionally, an ill-fitting denture will require a temporary reline. While lab-based relines are an alternative, the following products can be used for a chairside reline.

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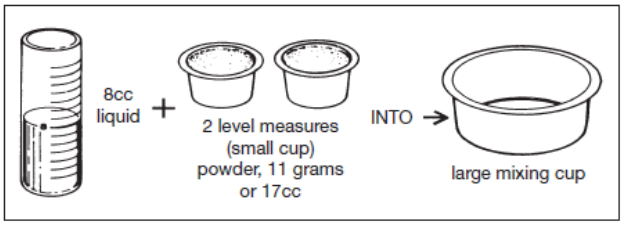
a. ***Coe-Soft*** *(GC America)*

- A temporary lining for acrylic dentures.
- While Coe-Comfort should be replaced every few days, Coe-Soft reline can last up to 3 months.



Instructions for Use

1. Remove undercuts in the denture, wash, and dry
2. Coat surfaces with Coe Lubricant to prevent unwanted adhesion
3. Pour 8mL of Coe-Soft Liquid into a mixing bowl and slowly add 11g of Coe-Soft Powder. Mix for 30 seconds
4. Spread the mixture into the denture and insert, having patient close lightly into occlusion. After 3 minutes, have patient move lips and cheeks to muscle trim
5. Remove denture, rinse under cold water, and trim excess
6. Re-seat denture and have patient FIRMLY close into occlusion. Hold for 5 minutes, remove denture, and rinse again
7. Use a hot spatula to smooth edges
8. Patient should NOT brush denture. Clean with cold water and wiping with wet cotton



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b. ***Visco-gel*** *(Dentsply Sirona)*

- A temporary soft lining for acrylic dentures.
- Self-curing, resilient methacrylate formulation



Instructions for Use

1. Remove pressure areas and undercuts in the denture, wash, and dry
2. Coat surfaces where material is not wanted with Separator to prevent unwanted adhesion
3. Pour the desired amount of material according to the ratio: 1 measure powder (3g) to 1 measure liquid (2mL). Mix for 30 seconds
4. Spread the mixture into the denture. Moistened fingers may help avoid initial material tackiness.
5. Insert after 2 or 3min from start of mix, remove excess, and have patient close lightly into occlusion for a minimum of 2 min. Patient should then move lips and cheeks to mimic functional movements for 2min
6. Remove denture after 7 or 8min from start of mix
7. Trim excess with a sharp instrument
8. Recall patient after a few days and re-evaluate

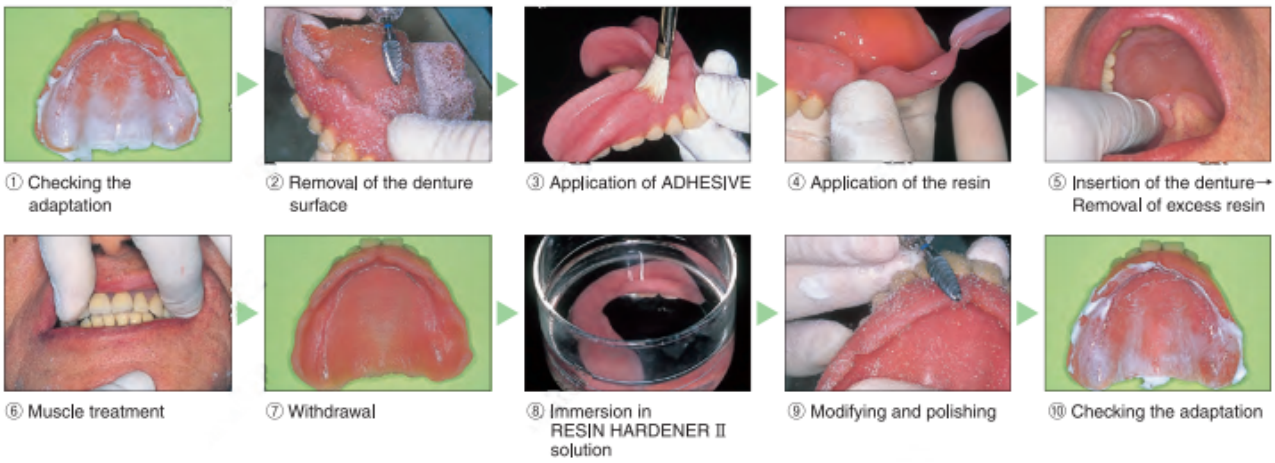
c. ***Rebase II Fast*** *(Tokuyama)*

- A chairside hard reline material



Instructions for Use

1. Remove the old resin and expose the new resin of the denture surface where it is necessary to be relined
2. Place separating agent on surfaces not to be added to
3. Apply adhesive in a single layer and leave for 20 seconds at room temperature
4. Measure one graduation of the powder for every two graduations of the liquid
5. Place the powder into the liquid and mix gently for 5 to 10 seconds
6. Apply resin within 20 to 60 seconds after mixing to all the surfaces being relined
7. Insert denture within 60 to 90 seconds after mixing has begun. Be cautious of excess
8. Perform appropriate treatment of the muscles in the mouth once the resin begins to harden
9. Trim as needed within 3 minutes from the start of mixing
10. Re-insert the denture in the patient's mouth for 3 to 5:30 minutes from the start of mixing
11. Once complete, resin hardener can be applied



Endodontic Materials

Patients will commonly run into trouble when cavities go untreated for too long. Root canal treatments happen on a regular basis; therefore, being well versed in the materials for them will help ease an often arduous appointment for the patients.

1. **Diagnosis and Instrumentation of Root Canals**
   - Endodontics involves the cleaning and shaping of a tooth's root canals before obturating them. Various products have been made to assist in this process. They work to help cut back the bacterial load chemically, along with helping to irrigate and lubricate the canals.

---

a. ***Endo-Ice*** *(Coltene)*
   - The normal response of nerve fibers can be tested by stimulating the nerve endings of the pulp with a cold spray. As a result of this test, the vitality (existing blood flow) of the tooth can be made



Instructions for Use

1. Decide which teeth to include for vitality testing, including control teeth
2. Spray a cotton applicator for a few seconds with Endo-Ice until saturated
3. Place on middle to cervical third of teeth to be tested and record stimulation and duration of response
4. Reapply Endo-Ice as needed throughout testing

---

b. ***OraSeal*** *(UltraDent)*
   - A caulking and putty material that effectively adheres to a wet dental dam to aid in isolation



Instructions for Use

1. Place a rubber dam around the desired teeth
2. Apply OraSeal material around the area of leakage
3. Shape as needed with a wet gloved finger, wet cotton swab, or instrument
4. Continue with procedure

---

c. ***EDTA Solution (17%)*** *(Pulpdent)*
   - A 17% solution of EDTA buffered to a neutral pH used as a calcium binding or "chelating" material to assist in the instrumentation of root canals
   - This solution helps decalcify the surface layer of dentin, making it easier to remove the dentin while enlarging and shaping the canal
   - Used as part of a final rinse protocol to remove the smear layer



Instructions for Use

1. Apply EDTA at entrance of canal with pellet or syringe
2. File/Ream canal in conventional manner
3. Reapply as necessary until the canal is enlarged to the desired size
4. Rinse the canal thoroughly to ensure all EDTA has been removed from the canal

---

d. ***Sodium Hypochlorite Dilution***
   - Dilution made to irrigate canals between files

Instructions for Use

1. Access the canals under rubber dam isolation
2. Remove chamber contents with bur and excavators
3. Flush with sodium hypochlorite
4. Continue to instrument the canal in conventional manner
5. Repeat: Sodium hypochlorite and RC-Prep can be alternated for the first two or three sizes of instruments, after which, only sodium hypochlorite should be used

e. ***RC-Prep Microdose Pipettes*** *(Premier Dental)*

- Glycol, urea peroxide, and EDTA in a water-soluble base
- Helps remove calcifications and lubricate the canal
- Will form bubbles when reacting with sodium hypochlorite



Instructions for Use

1. Access the canals under rubber dam isolation
2. Remove chamber contents with bur and excavators, and flush with sodium hypochlorite
3. Introduce RC-Prep into the canal using the microdose pipette
4. Continue to instrument the canal in conventional manner
5. Repeat: Sodium hypochlorite and RC-Prep can be alternated for the first two or three sizes of instruments, after which, only sodium hypochlorite should be used

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f. ***CHX-Plus*** *(Vista Dental Products)*

- Advertised as a 2% Chlorhexidine Gluconate Solution plus powerful wetting agents and proprietary surface modifiers that improve the depth of penetration than normal 2% Chlorhexidine
- *Do NOT use this material when canal is wet with sodium hypochlorite, as a brown precipitate will form*



Instructions for Use

1. After cleaning and shaping has been performed using EDTA (NOT NaOCl), apply CHX-Plus at entrance of canal with pellet or pipette
2. A file may be used to help the material into the canal
3. Rinse the canal thoroughly and dry using paper points before obturating

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g. ***Formocresol Bottle*** *(Pearson)*

- 48.5% formaldehyde, 48.5% creasol, and 3% glycerine
- Powerful medicament for root canal therapy
- Will more commonly be used for pediatric pulp devitilization
- Should this material be needed, only use under the supervision of appropriate faculty

2. Obturation and Sealing

- After cleaning and shaping is complete, the canal must be filled. Western University of Health Sciences utilizes gutta percha for this; however, there are various sealers that can be used depending on each situation.

---

a. ***EndoSequence BC Sealer*** *(Brasseler)*

- Calcium silicate composition which *requires* the presence of water to set and harden
- Premixed injectable bioceramic paste for permanent root canal filling and sealing applications
- Insoluble, radiopaque, aluminum-free
- Used for permanent obturation of the root canal
- Setting time is 4 hours; however, without moisture it can be more than 10 hours



Instructions for Use

1. Thoroughly prepare and irrigate the root canal using standard methods
2. Dry canals using paper points
3. Two different methods to use material:
   a. *Traditional Method*: Coat gutta percha or hand file with sealer and use to apply to sealer to the canal walls
   b. *Tip Delivery Method*: Remove cap and secure the BC Tip. Insert tip into the canal no deeper than coronal 1/3rd. Gently and smoothly dispense a small amount into the canal, then follow with a 15 hand file to lightly coat the canal walls. Then, coat the master cone and insert into the canal
4. If desired, place additional points into canal and continue with the obturation using standard techniques

---

b. ***ThermaSeal Plus Ribbon*** *(Dentsply)*

- Two-component paste root canal sealer based on epoxy-amine resin
- Advertised as offering long-term seal with outstanding dimensional stability, self-adhesive properties, and a high radiopacity
- Indicated for use in permanent obturation of root canals
- With two components, remember to bleed the syringe prior to using



Instructions for Use

1. This material will be used identically to the BC Sealer. Refer above for instructions for use

3. Continuing Root Canal Therapy

- When RCT cannot be completed in a single appointment, some materials can be used to protect the tooth in the window between appointments.
- There are other materials used to finish the RCT by sealing the orifice after obturation.

---

a. ***UltraCal XS*** *(Brasseler)*

- 34% Calcium Hydroxide Paste
- Advertised as radiopaque material with a high pH (12.5)
- Indications for use include pulp capping, lining pulpotomies, or as a temporary dressing



Instructions for Use

1. Instrument a canal either partially or fully, rinse, and dry
2. Remove the cap and apply the NaviTip to the syringe
3. Insert the syringe into the canal passively, without binding
4. Fill the canal from apical portion outward staying at least 2mm shy of apex. A lentulo spiral or file can be used to help carry the material towards the apex
5. Continue with temporary restoration

---

b. ***Dyractflow*** *(Dentsply)*

- Fluid compomer
- Advertised for offering long-term fluoride release
- Available in shades A2, A3, A4, B1, and Opaque
- WesternU will primarily use this product to cover the orifice after obturation



Instructions for Use

1. Apply appropriate adhesive system
2. Dispense Dyractflow either onto an instrument or directly into the preparation with gentle, even pressure, beginning with the deepest area first
   a. Avoid lifting the tip to minimize air entrapment
   b. Do not leave too long under intensive light or the material will begin to cure
   c. Fill in 2mm increments
   d. When finished, wipe tip against cavity wall as the tip is removed
3. Light cure each increment for 20 seconds

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c. ***EndoSequence BC Liner*** *(Brasseler)*

- Advertised as a first-generation bioactive hybrid resin optimized for use as a liner
- Properties similar to an RMGI
- WesternU will primarily use this product to cover the orifice after obturation



Instructions for Use

1. After appropriate instrumentation and obturation of a root canal under rubber dam isolation, place any necessary matrices for proper restoration
2. Bleed syringe to ensure equal flow and attach mixing tip with appropriate dispenser
3. Gently dry area surrounding orifice(s) but do not desiccate
4. Dispense lining material over the orifice(s) to create a coronal seal no more than 2 mm in depth
5. Allow the material to self-cure for 20-30 seconds before light curing for 20 seconds
6. Continue restoring with desired core build-up material

4. Repairs

- Occasionally, there will be a need to repair and seal a part of a tooth.

---

a. ***EndoSequence BC RRM Putty*** *(Brasseler)*

- Calcium silicate composition which *requires* the presence of water to set and harden
- Premixed injectable bioceramic paste
- Insoluble, radiopaque, aluminum-free
- Intended for pulp capping and root repairs
- 2 hour minimal set time. If using Fast set, 20 minute set time
- Store in a foil pouch in a dry area at room temperature



Instructions for Use

1. Thoroughly prepare and irrigate the root canal using standard endodontic techniques
2. Unscrew the jar, remove the desired amount, and re-cap the jar immediately
3. Use a sterile instrument to apply the material to the intended anatomic section of the root canal and compress with a plastic instrument
4. Remove excess

---

b. ***ProRoot MTA (Mineral Trioxide Aggregate)*** *(Dentsply)*

- Calcium silicate composition
- Fine hydrophilic particles that set in the presence of water
- Material will set over a 3 hour period. Should the material need to be set before proceeding with a procedure, wait at least 4 hours
- Intended for pulp capping, root repairs, pulpotomies on primary teeth, and other endodontic uses



Instructions for Use

1. Prepare tooth as needed
2. Open pouch of powder onto a mixing pad
3. Squeeze micro-dose ampoule onto the mixing pad next to the powder
4. Slowly incorporate the liquid gradually into the powder until all powder has been hydrated
5. Place material in desired area
6. Continue with procedure

5. Walking Bleach Method

- Root canal therapy occasionally has the effect of graying a tooth. When this happens in an esthetic area, one option to reverse it is to try the "Walking bleach method" to internally bleach the tooth.

---

a. ***Superoxol/Sodium Perborate*** *(Sultan)*

- Oxidizing materials that are used together to bleach endodontically treated anterior teeth
- If the tooth in question has a large restoration, a restorative solution should be considered



Instructions for Use

1. Open the endodontic access under rubber dam isolation
2. Remove gutta percha, filling material, and sealer to the level of the facial gingiva
3. Seal the opening of each root canal with composite or RMGI to prevent the bleaching material from entering the canals to the level of the CEJ
4. Dispense a small amount of sodium perborate into a dappen dish. Add Superoxol dropwise while stirring until a moderately stiff paste is formed
5. Place the paste in the pulp chamber
6. Seal the access hole with a temporary filling material
7. Evaluate the condition and color of the tooth in 3 to 5 days from initial placement
8. Repeat as necessary until desired color is achieved
9. Remove the paste, rinse the pulp chamber, and restore the access opening

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6. Canal Retreatment

- Despite a high rate of success, there are times when a root canal must be reattempted. During a retreatment, one material can be provided to help remove the gutta percha.

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a. ***Chloroform*** *(Sultan)*

- This product is a general solvent that can help clean up hardened cements and materials
- CAUTION: This material is a poison that can be fatal if swallowed. The vapor can also be harmful. Always use a rubber dam, mask, and suction when working with this product
- Should this material be needed, only use under the supervision of appropriate faculty

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7. Caries Indicator Dye

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a. ***To Dye For Caries Indicator Blue*** *(Roydent)*

- Advertised as a root canal locator, caries indicator, and dental crack locator
- Research does not fully support the use of a caries indicator; however, endodontists do use this material for assistance with finding canals and fractures. This will be the primary use of this material here at WesternU



Instructions for Use

1. Cut the end of the pipette using scissors
2. Coat the surface to be examined. After 10 seconds, rinse and dry
3. Remove caries slowly with a bur or hand excavator
4. Repeat as needed until all carious material is removed
5. Restore with material of choice

Miscellaneous Materials

1. Polishing Medium

a. ***Pumice Preppies*** *(Whip Mix)*

- Flour of pumice paste for multiple dental applications
- Cleaning agent that contains no flavoring agents, oils, or fluoride
- This product can be useful for cleaning a surface prior to acid etching or restoration cementation, polishing a restoration, prophylaxis of heavily stained teeth, and more



Instructions for Use

1. Using a prophy angle, fill the cup with the paste while running the angle at low speed
2. Apply paste gently to teeth
3. Throughout procedure, sufficient paste should be maintained on the cup
4. Using steady, slow speed, flare the cup during engine polishing to ensure full coverage
5. Saliva evacuation is recommended to avoid excessive dilution and swallowing of the paste
6. Once complete, rinse well and continue with procedure

2. Band Cementation

a. ***GC Fuji ORTHO Band*** *(GC America)*

- Blue, resin modified glass ionomer (RMGI) for bonding orthodontic bands
- Mixes within 15sec and has a 3min 30sec working time with a final set after 3min in the mouth



Instructions for Use

1. Adapt an appropriate band to the desired tooth and clean the band of any contaminants
2. Clean the tooth with pumice, rinse, and dry. Do not desiccate
3. Load the cartridge into the Paste Pak Dispenser
4. Slide the release lever forward until it stops
5. Remove the cartridge cover and bleed a small amount of material
6. Depress the lever to dispense the required amount of pastes onto the mixing pad
7. Mix thoroughly for 15 to 20sec
8. Using a suitable instrument, place the material throughout the band
9. Seat the band
10. Once in the final position, clean excess cement after initial set

3. Post-Surgery Medicaments

a. ***Dry Socket Paste*** *(Sultan)*

- Treats alveolar osteitis (“dry socket syndrome”) and provides immediate pain relief
- Contains 4% guaicol, balsam peru, 4% eugenol, and 1.6% chlorobutanol



Instructions for Use

1. Establish a diagnosis of alveolar osteitis; typically characterized as a constant dull or throbbing ache, exacerbated by touch, approximately 2 days post extraction. No clot will be present, and food debris will usually be in the socket
2. Rinse and de-bride the socket
3. Use a syringe or flat-bladed instrument to place the Dry Socket Paste into the socket
4. Tamp down the paste ensuring coverage of all exposed bone
5. Do not cover the paste
6. Instruct patient that another treatment may be necessary in 3 to 5 days depending on symptoms and healing. There is no need to appoint for removal of material

4. General Purpose Acrylic

- These materials are largely available in the lab for use with prosthodontic faculty oversite.

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a. ***Pattern Resin*** *(GC America)*

- An acrylic resin that can be used for making patterns for post and core build-ups to fixation prior to soldering
- Powder: Polymethylmethacrylate, Polyethylmethacrylate, Dibenzoyl peroxide
- Liquid: Methylmethacrylate, 2-Hydroxyethyl-Methacrylate
- While this product has many uses, a general procedural use will be listed below. See the manufacturer's guide for specific procedural needs



Instructions for Use

1. Dispense an adequate amount of powder and liquid into individual cups
2. Moisten a brush with the liquid
3. Pick up a small amount of powder with the moistened brush
4. With a bead on the brush, the resin is ready for use
5. Apply as needed and repeat process until desired design is complete
6. Material will polymerize with time

---

b. ***Jet Acrylic Liquid and Powder*** *(Lang Dental)*

- Fast-curing provisional crown and bridge material
- Methyl methacrylate monomer with a methyl methacrylate polymer



Instructions for Use

1. Shake the liquid monomer before use
2. In the matrix, moisten the surface of the area to be provisionalized with the liquid monomer
3. Spray Jet powder from the dispenser bottle
4. Alternate liquid and powder until the matrix is filled, ending with a liquid layer
5. While the acrylic reaches a paste like consistency, coat the prepared teeth with a silicone lubricant
6. Once a paste consistency is reached, seat the matrix intraorally and hold in place for 3min
7. Remove and trim excess. Do NOT over trim
8. Reseat matrix until the material reaches a rubbery state and cannot be distorted
   - Do NOT allow material to get hot in the mouth
9. Remove matrix and temporary. Allow 10 more minutes of curing
10. Remove provisional from matrix and check fit
11. Trim and polish

---

c. ***Hygenic Repair Material Perm Reline Repair Resin*** *(Coltene)*

- Denture acrylic material



Instructions for Use

1. Mix 3 parts powder to 1 part liquid by volume and knead to a smooth dough
2. Pack material when it is no longer sticky, but before it begins to feel dry and grainy
3. Material can be worked for about 5 minutes at room temperature
4. Leave in final closure under pressure for 1 hour to ensure a more perfect cure

5. Minor Gingival Retratction

---

a. ***Ultrapak*** *(Ultradent)*

- 100% knitted cotton cord
- Available in size 000, 00, 0, 1, 2, and 3
- Used in areas where minor tissue displacement is needed, such as a subgingival class V preparation or a crown impression
- Optimal tissue displacement occurs within 3-8 minutes



Instructions for Use

1. Prepare tooth and achieve hemostasis
2. Once ready for an impression or restoration, cut the length of cords adequate for the case. The first cord will often be thinner than the second when being used for an impression
3. For 2 cord technique when taking final impressions of crowns:
   a. Begin to place the first cord, dry, into the sulcus. Start at a line angle for ease, but end at a different location
   b. When placing a cord, use a cord packer to gently pack into what was previously placed
   c. Next place a second cord, wet. Either with a hemostatic agent or with water so it will be more easily removed and less traumatic to the tissue
   d. Allow cords to remain in the sulcus for 3 to 8 minutes until adequate tissue displacement has occurred
   e. When ready to take the impression, remove the top cord, but leave the original cord in place. Proceed with impression
   f. After the impression, ensure the final cord is removed as well

Laboratory Materials

Despite much of dentistry being chairside, every dentist should have knowledge of the laboratory materials that make their profession possible.

1. **Stone**
   - Dentistry uses various types of stone throughout the lab work to achieve a variety of goals.
   - Here is a quick guide to the ISO guidelines for the different types of gypsum:

ISO 6873 Dental Gypsum Specification

Type	Setting Expansion (%)	Compressive Strength (psi)
1	0 - 0.15%	580 – 1,160 psi
2	0 - 0.30%	1,300 psi (minimum)
3	0 - 0.20%	2,900 psi (minimum)
4	0 - 0.15%	5,100 psi (minimum)
5	0.16 - 0.30%	5,100 psi (minimum)

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a. ***Mounting Stone White: Fast Set*** *(Whip Mix)*
   - ISO Type 3
   - Low-expansion, fast-setting stone for mounting casts to articulators

Physical Properties

- Water/Powder Ratio **26mL/100g**
- Working Time **2-3 minutes**
- Setting Time **5 minutes**
- Setting Expansion 0.08%
- Early Compressive Strength, Wet (1 hour) 4,600 psi (32 MPa)
- Early Compressive Strength, Dry (48 hour) 8,500 psi (59 MPa)

Possible Uses

1. Mounting casts to articulators



b. ***Microstone*** *(Whip Mix)*
- ISO Type 3
- Microstone is a fine particle size, dense surface stone used for all general laboratory purposes

Physical Properties

• Water/Powder Ratio	**28mL/100g**
• Working Time	**5-7 minutes**
• Setting Time	**13 minutes**
• Setting Expansion	0.12%
• Early Compressive Strength, Wet (1 hour)	4,500 psi (31 MPa)
• Early Compressive Strength, Dry (48 hour)	8,500 psi (59 MPa)

Possible Uses

1. Diagnostic impressions
2. Casts for pull-down appliances
3. Custom Trays
4. Removable final casts



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c. ***Die Keen Green*** *(Kulzer)*
- ISO Type 5
- High-strength with low setting expansion die stone

Physical Properties

• Water/Powder Ratio	**21mL/100g**
• Working Time	**5-7 minutes**
• Setting Time	**10-13 minutes**
• Setting Expansion	0.16-0.30%
• Early Compressive Strength, Wet (1 hour)	5080 psi (≥35 MPa)
• Early Compressive Strength, Dry (48 hour)	Not Available

Possible Uses

1. Final fixed prosthetic casts



2. Tray Materials

- Whether for custom trays or for trial bases, these materials will help create a custom fit for each patient.

---

a. ***Triad TruTray*** *(Dentsply Sirona)*

- A translucent, uniform sheet of material that can be adapted to a cast for custom tray fabrication



Instructions for Use

1. Block out any undercuts on the cast
2. If desired, place baseplate wax for tissue relief
3. Apply a releasing agent/lubricant
4. Remove Triad TruTray material from the light-proof packaging and adapt it to the cast
5. Trim excess with a sharp blade. Excess can be used to create a handle
6. Paint tray with an air barrier coating
7. Place cast and tray on the turntable of the Triad unit
8. Cure for 2min
9. Once fully cured, remove the air barrier coating with soap and water, and adjust and finish the tray





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b. ***Triad Denture Base*** *(Dentsply Sirona)*

- Advertised as a versatile material that can be used for denture base fabrication, relines, repairs, orthodontic, appliances, custom trays, implant applications, etc.
- It will primarily be used to create a record base and for denture repairs. Instructions for creating a record base are below:



Instructions for Use

1. Block out any undercuts on the cast
2. Apply a releasing agent/lubricant
3. Remove Triad material from the light-proof packaging and adapt it to the cast
4. Trim excess with a sharp blade. Do not over thin the material.
5. Place cast and tray on the turntable of the Triad unit
6. Cure for 2min
7. Separate tray from cast, apply an air barrier coating to the entire surface of the record base, and cure again for 6min
8. Once fully cured, adjust and finish the tray. Carbide burs are recommended

3. Miscellaneous Laboratory Materials

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a. ***Lab Putty and Accelerator*** *(Coltene)*

- Polysiloxane laboratory modeling material



Physical Properties

Mixing Time	30-45 seconds
Working Time	90 seconds
Setting Time	5-7 seconds

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b. ***Gi-Mask Automix*** *(Coltene/Whaledent)*

- Medium-viscosity silicone for the elaboration of gingival masks
- In particular, this material is commonly used to replicate the gingiva around an implant impression
- This product has a 2min working time with a 7min setting time



Instructions for Use

1. Insert the cartridge into the dispenser
2. Remove the cartiridge cap and bleed a small amount from the cartridge by squeezing the trigger. Remove any contaminants with a towel
3. Attach the mixing tip, followed by the applicator tip
4. Eject the material by pulling the trigger and instrument to desired location and shape
5. Allow the material to set for 7 minutes
6. Continue with creating the cast. The mixing tip can be left on the cartridge to seal it until it is to be used again. This will help prevent cross contamination

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c. ***LC Block-Out Resin*** *(Ultradent)*

- The LC Block-Out Resin is a resin designed to provide reservoir space for bleaching trays
- The space should be 0.5mm thick on the labial surface. It should be 1.5mm from the gingival line and NOT extend onto the incisal/occlusal surfaces. Light cure resin for 20-40 seconds

